Competing Risks and Multi-State Models

Concepts, methods and software

Course given at *Karolinska Institute*

May 5-8, 2026

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Outline

Day 1; Competing risks Examples. The main quantities. Data structure. Estimation: cause-specific hazard, multi-state approach

Day 2; Regression Estimation: subdistribution hazard. Regression models. Type of study questions: inferential, predictive, causal. Which model to use?

Day 3; Multi-state models Objectives. Terminology. Data. Non-parametric estimation.

Day 4; Prediction; Interpretation Prediction in multi-state models. Interpretation; when to choose which quantity?

Part II

Estimation: Multi-State Approach

Contents:

1. Cause-specific hazard
2. Aalen-Johansen estimator
3. Display formats
4. Independent censoring?

Outline

Aalen-Johansen estimator
- Confidence intervals

Display formats

Marginal versus Cause-specific
- All quantities and their estimators
- Assumptions on censoring and delayed entry
- Independent right censoring?
- What can we do?

Regression; proportional cause-specific hazards
- Standard Cox model
- Proportional cause-specific hazards

Probabilistic setup data

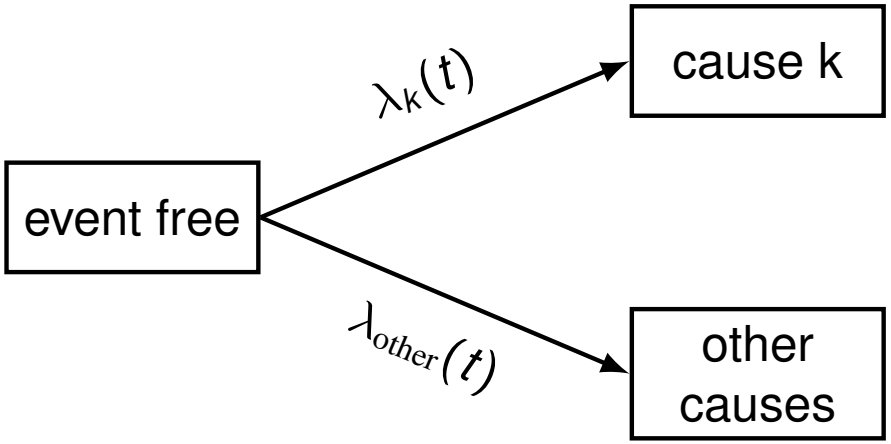
- Event time T_i; may be right censored (administratively, loss to follow-up)
  Censoring time C_i
  $X_i = \min\{T_i, C_i\}$, $\Delta_i = I(T_i \leq C_i)$
- Left truncation (late entry).
  Entry time V_i
- Sample: $\{V_i, X_i, E_i\Delta_i\}, i = 1, \ldots, N$
- Assume (V_i, C_i) independent of/non-informative for (T_i, E_i) (possibly conditional on covariables in regression model)

Observed data

$\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$
Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$

Cause-specific hazard



- Individuals with a competing event are no longer at risk $\Longrightarrow$ leave the risk set

$$\widehat{\lambda_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r(t_{(i)})}$$

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- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
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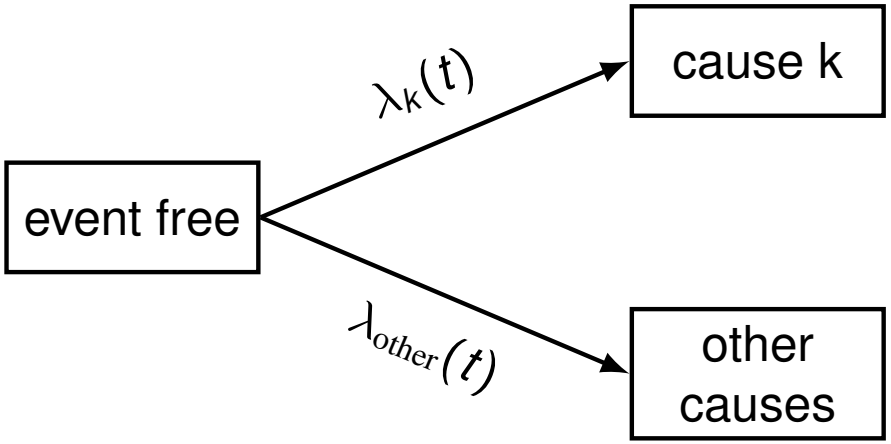
- $r(t_{(i)})$ number observed at risk

Cause-specific hazard



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- Standard rate estimate (hazard with single event type). Same estimator as for marginal hazard (assuming independence), but different interpretation, unless censoring due to competing risks is non-informative

Aalen-Johansen estimator of F_k

- Plug-in estimator based on $F_k(t) = \int_0^t \mathrm{P}\{T \geq s\}\lambda_k(s)ds$:

$$\widehat{F_k}^{\mathrm{AJ}}(t) = \sum_{i:t_{(i)} \leq t} \widehat{F}^{\mathrm{PL}}(t_{(i)}-) \times \widehat{\lambda_k}(t_{(i)}) \text{ with}$$

$$\widehat{\lambda_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r(t_{(i)})} \qquad \text{cause specific hazard}$$

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Observed data

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- Single event type: equal to Kaplan-Meier
- Competing risks: same hazard estimate as for marginal distribution, but cumulative quantity different

Example calculations

- 115 individuals die of A (60 at 2 years, 55 at 6 years)
- 85 individuals die of B (40 at 4 years, 45 at 8 years)

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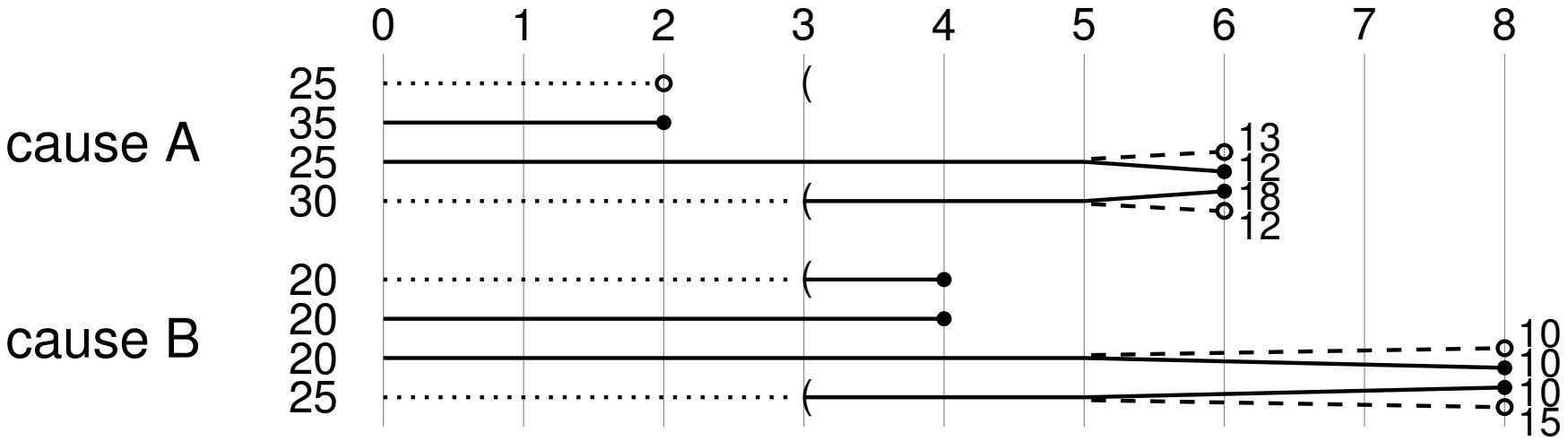
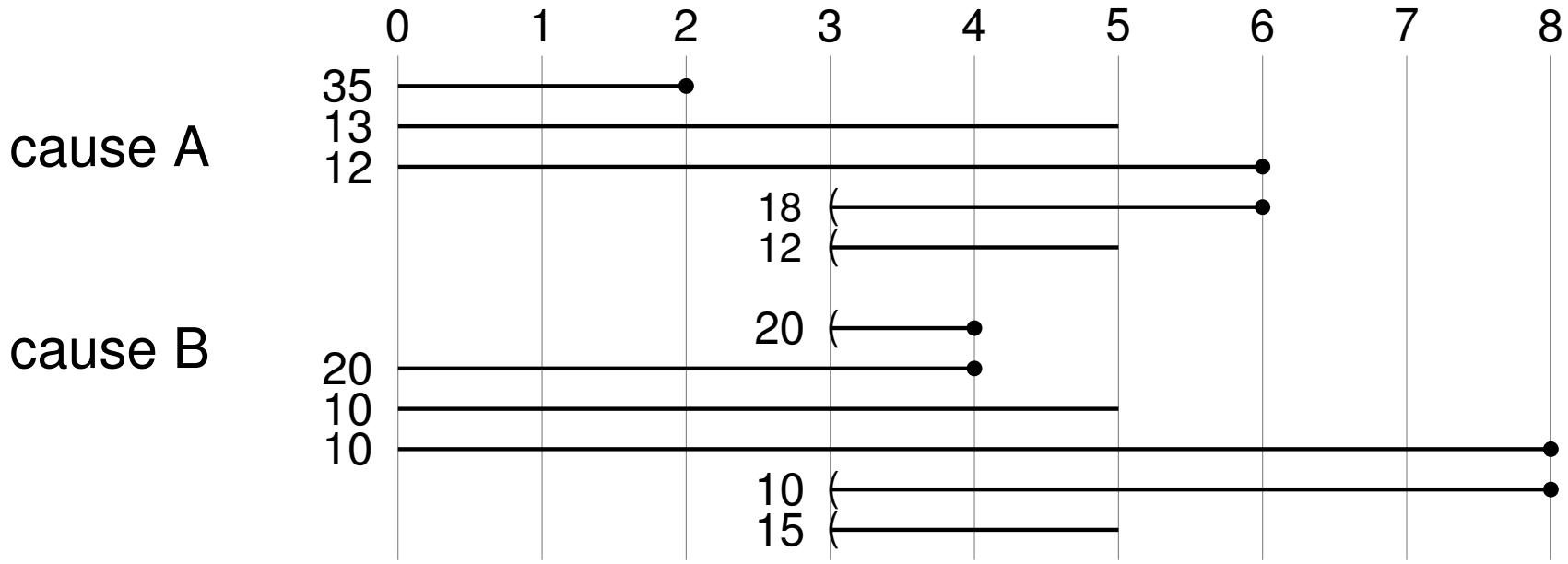


- 50 individuals censored after 5 years
- 100 individuals enter 3 years after the time origin (if they are still event free)

Detailed calculation of $\widehat{F_A}^{\mathrm{AJ}}(2)$ and $\widehat{F_A}^{\mathrm{AJ}}(6)$



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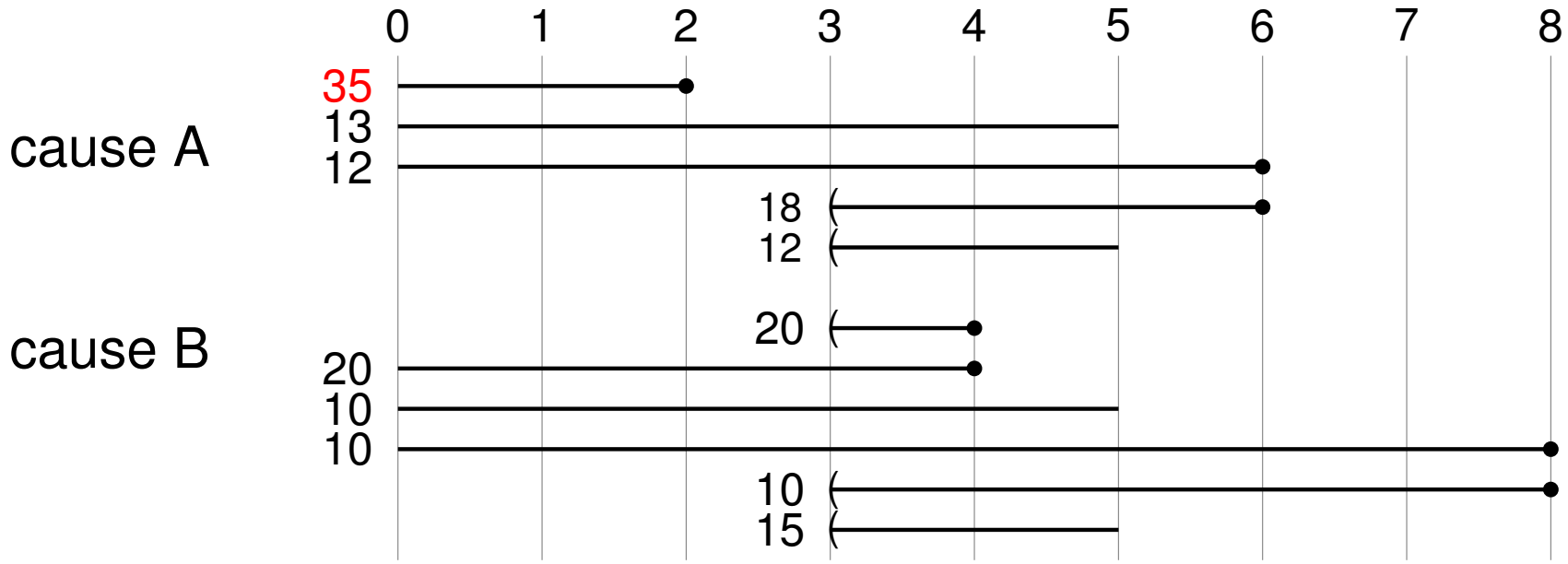

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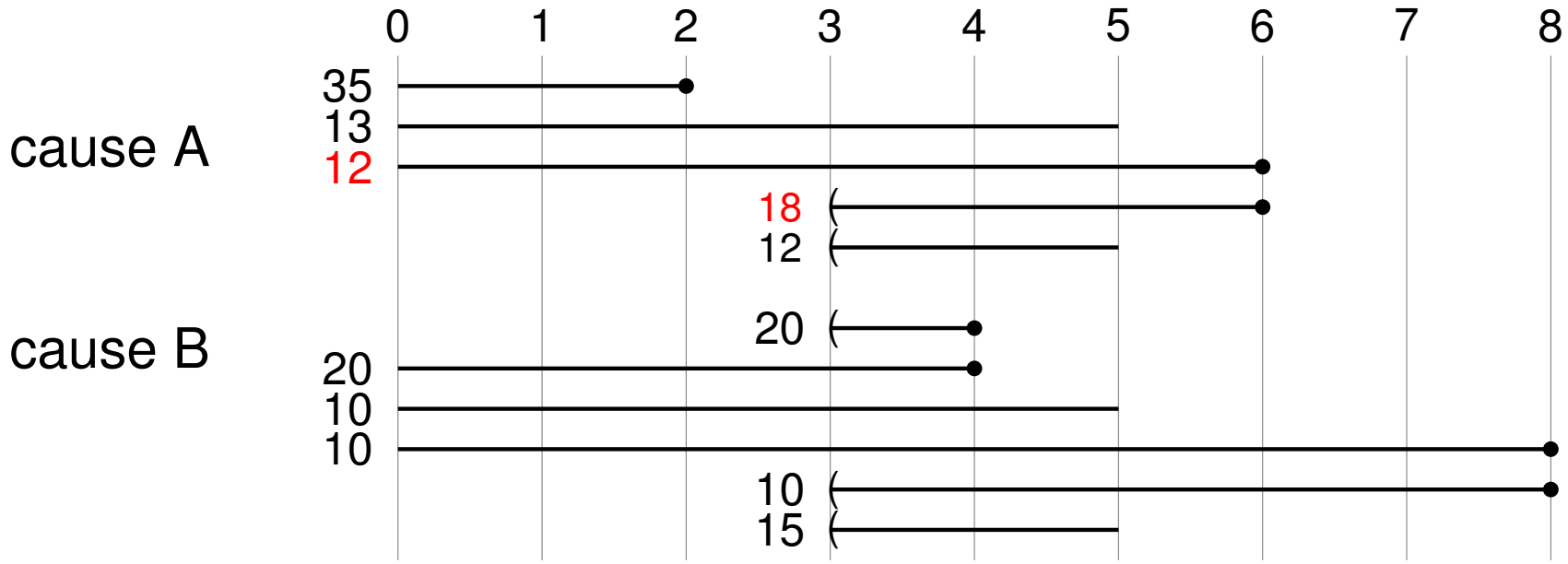

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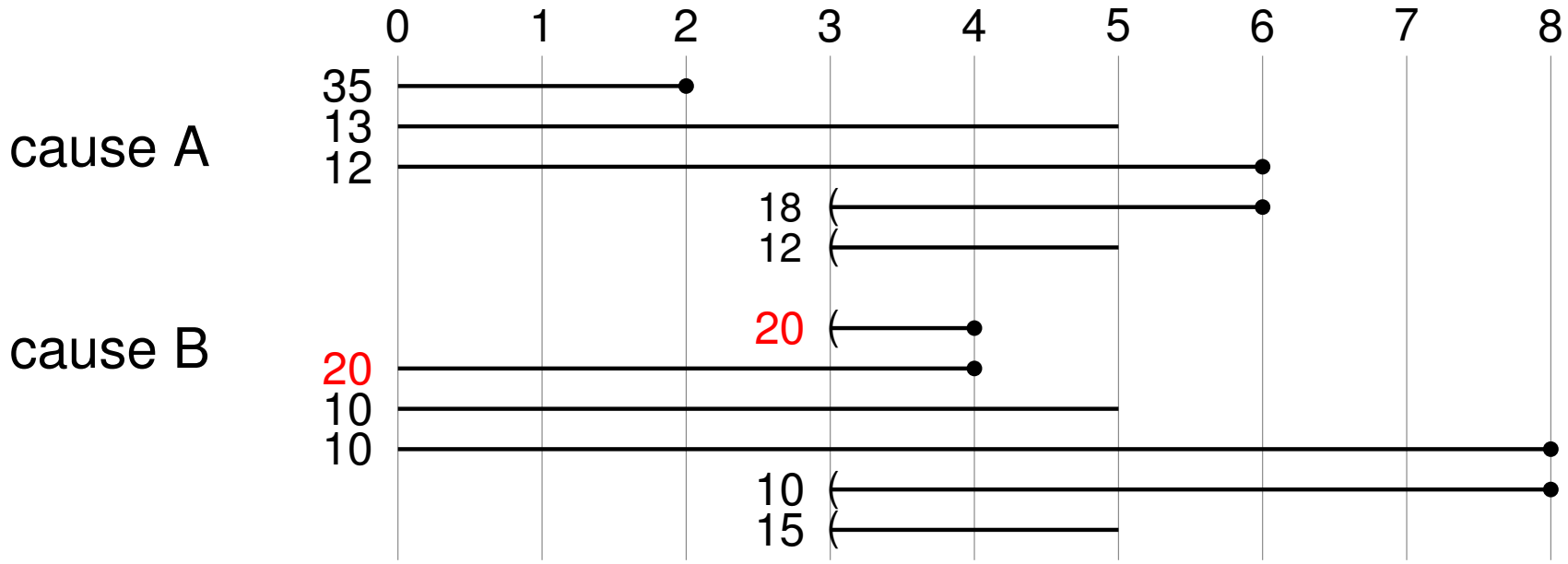

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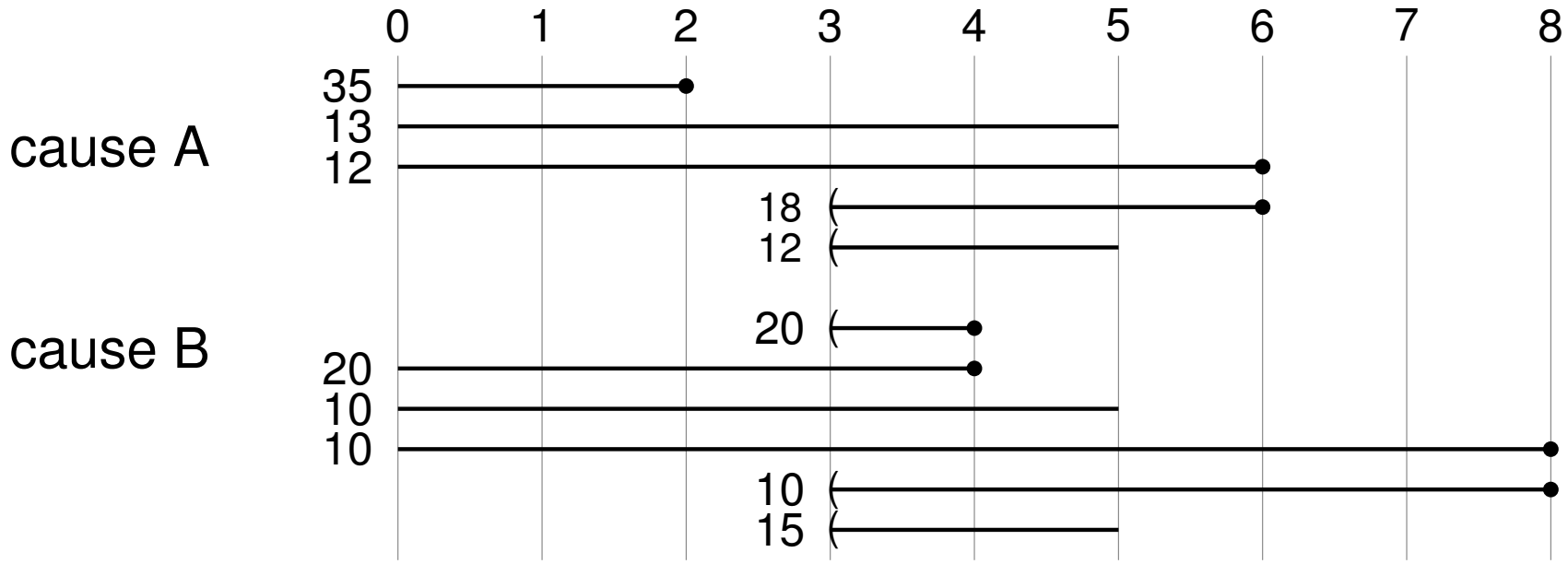

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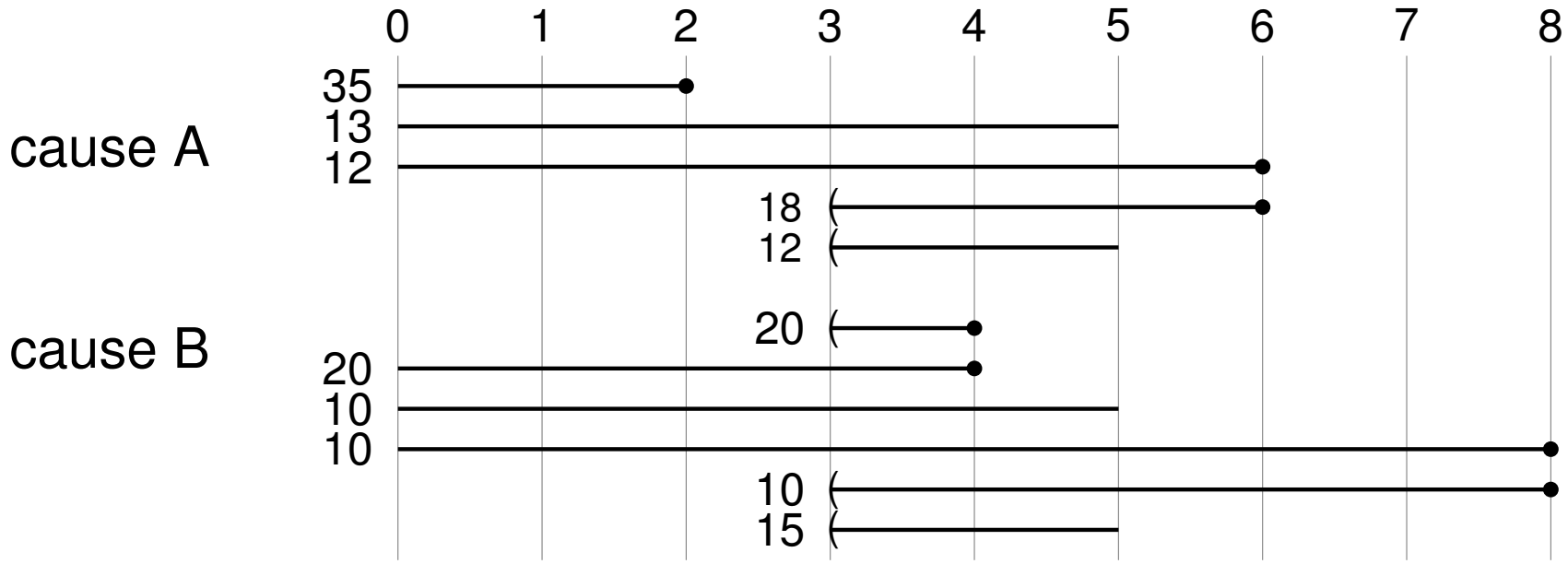

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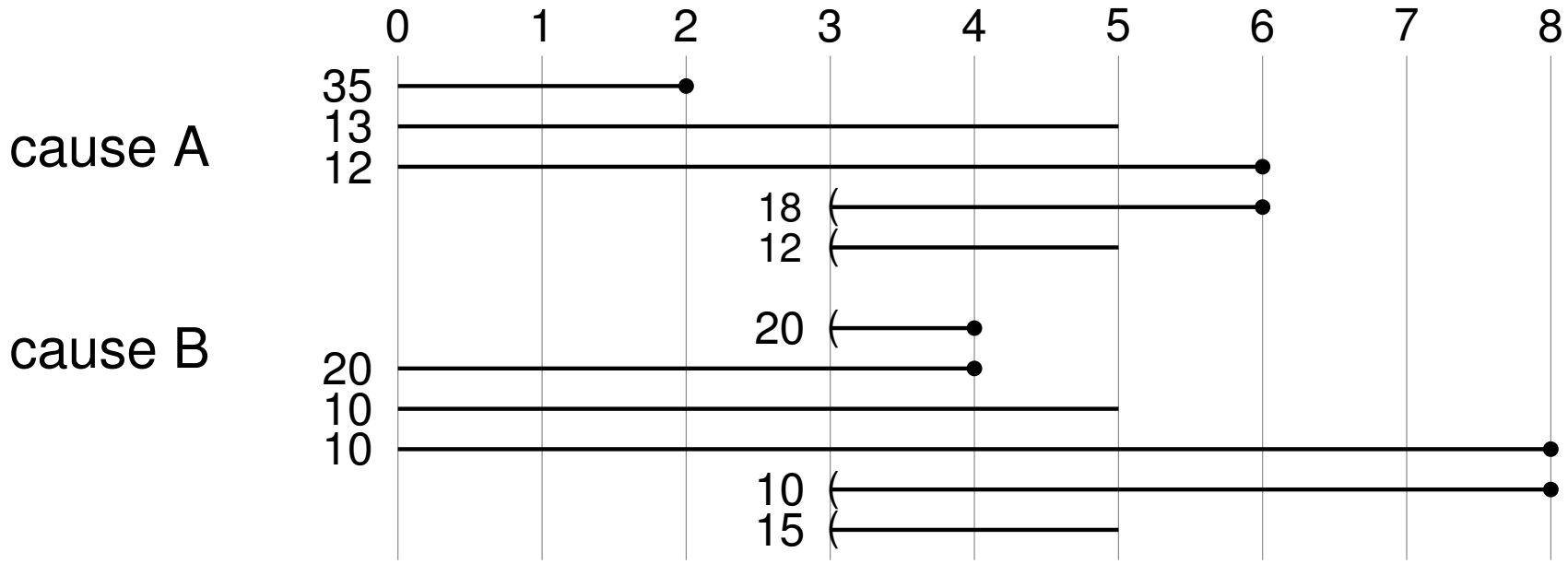

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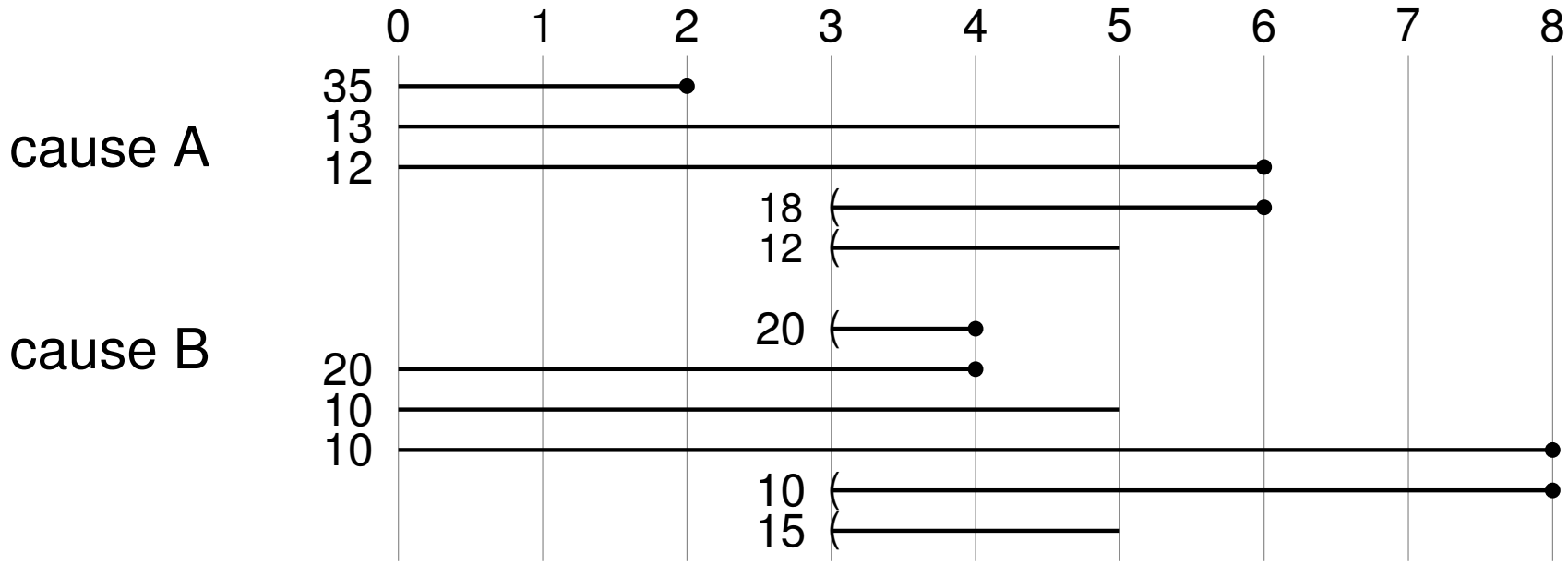

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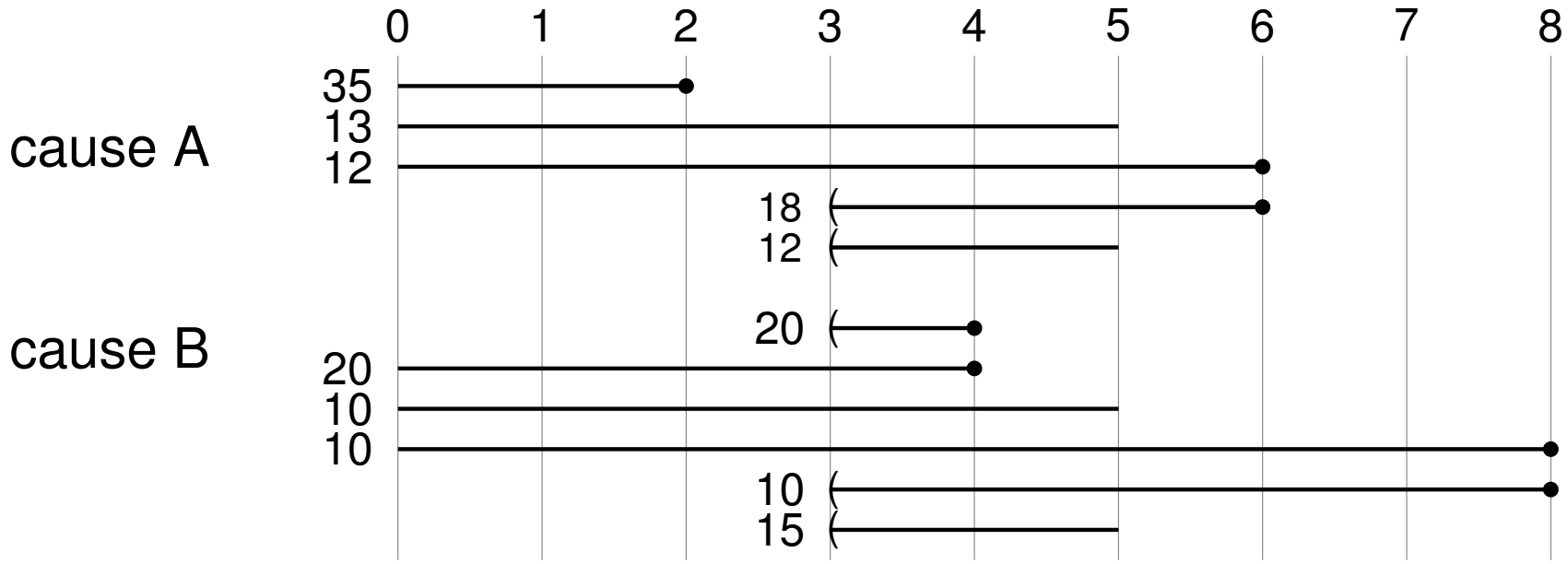

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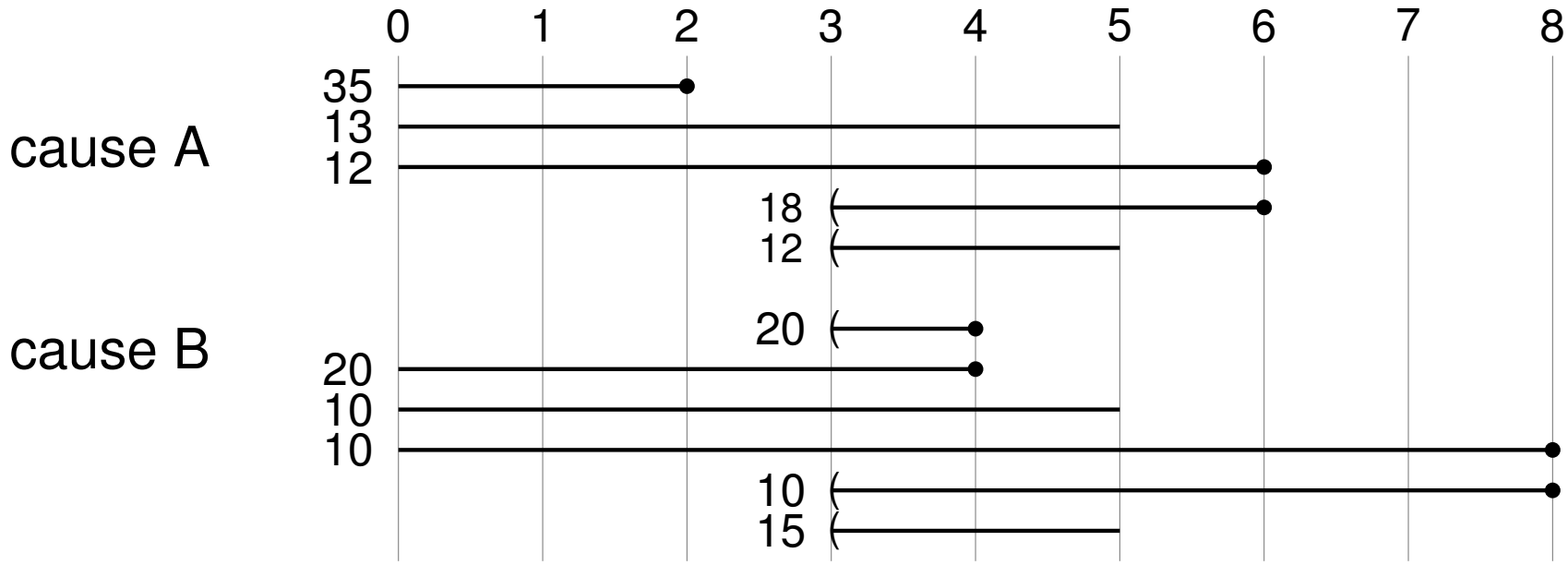

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Confidence intervals, Kaplan-Meier

Start with variance of Nelson-Aalen estimator $\widehat{H}(t)$

$$\text{Aalen:} \quad \widehat{\text{var}}\left\{\widehat{H}(t)\right\} = \sum_{t_{(j)} \leq t} \frac{d(t_{(j)})}{r^2(t_{(j)})} \tag{1}$$

$$\text{Greenwood:} \quad \widehat{\text{var}}\left\{\widehat{H}(t)\right\} = \sum_{t_{(j)} \leq t} \frac{d(t_{(j)})}{r(t_{(j)})\{r(t_{(j)}) - d(t_{(j)})\}} \tag{2}$$

Use the delta method and $\widehat{\overline{F}}(t) \approx \exp\{-\widehat{H}(t)\}$ to obtain

$$\widehat{\text{var}}\left\{\widehat{\overline{F}}(t)\right\} = \left\{\widehat{\overline{F}}(t)\right\}^2 \times \widehat{\text{var}}\left\{\widehat{H}(t)\right\}. \tag{3}$$

Confidence intervals, choice in scale

- On "plain" or "linear" scale:

$$\widehat{\overline{F}}(t) \pm k_\alpha \, \widehat{\text{s.e.}}\left\{\widehat{\overline{F}}(t)\right\} = \widehat{\overline{F}}(t) \pm k_\alpha \, \widehat{\overline{F}}(t) \times \widehat{\text{s.e.}}\left\{\widehat{H}(t)\right\}$$

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- On "log scale" [log survival; use $\log\{\widehat{\overline{F}}(t)\} \approx -\widehat{H}(t)$]:

$$\exp\left[-\widehat{H}(t) \pm k_\alpha \, \widehat{\text{s.e.}}\{\widehat{H}(t)\}\right] = \widehat{\overline{F}}(t) \times \exp\left\{\frac{\pm k_\alpha \, \widehat{\text{s.e.}}\{\widehat{\overline{F}}(t)\}}{\widehat{\overline{F}}(t)}\right\} \tag{4}$$

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- "log-log" or "complementary log-log" scale (use $\log \widehat{H}(t) \approx \log[-\log\{\widehat{\overline{F}}(t)\}]$)

The "log" scale has best overall performance, but "log-log" scale best for small sample sizes (≈ 25, $\leq 50\%$ censoring)

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- Transformation to $\widehat{\mathrm{var}}\left\{\widehat{F_k}(t)\right\}$. Complicated formula
- Same scales as in classical setting, obtained directly via expression in $\widehat{\overline{F_k}}$; e.g. "log" scale based on

$$\widehat{\overline{F_k}}(t) \times \exp\left\{\frac{\pm k_\alpha \, \widehat{\mathrm{s.e.}}\{\widehat{\overline{F_k}}(t)\}}{\widehat{\overline{F_k}}(t)}\right\},$$

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- On "log scale" [log survival; use $\log\{\widehat{\overline{F}}(t)\} \approx -\widehat{H}(t)$]:

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- "log-log" or "complementary log-log" scale (use $\log \widehat{H}(t) \approx \log[-\log\{\widehat{\overline{F}}(t)\}]$)

The "log" scale has best overall performance, but "log-log" scale best for small sample sizes (≈ 25, $\leq 50\%$ censoring)

Confidence intervals, competing risks

- Based on estimator of $\mathrm{var}\{\widehat{\Lambda_k}(t)\}$, same as in classical setting
- Transformation to $\widehat{\mathrm{var}}\left\{\widehat{F_k}(t)\right\}$. Complicated formula
- Same scales as in classical setting, obtained directly via expression in $\widehat{\overline{F_k}}$; e.g. "log" scale based on

$$\widehat{\overline{F_k}}(t) \times \exp\left\{\frac{\pm k_\alpha \, \widehat{\mathrm{s.e.}}\{\widehat{\overline{F_k}}(t)\}}{\widehat{\overline{F_k}}(t)}\right\},$$

- CI around $\widehat{\overline{F_k}}$ can be transformed to CI around $\widehat{F_k}$. There is also a direct approach, in which $\overline{F_k}$ in above formula is replaced by F_k.

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Naming of scales inconsistent

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- Confidence intervals

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- Independent right censoring?
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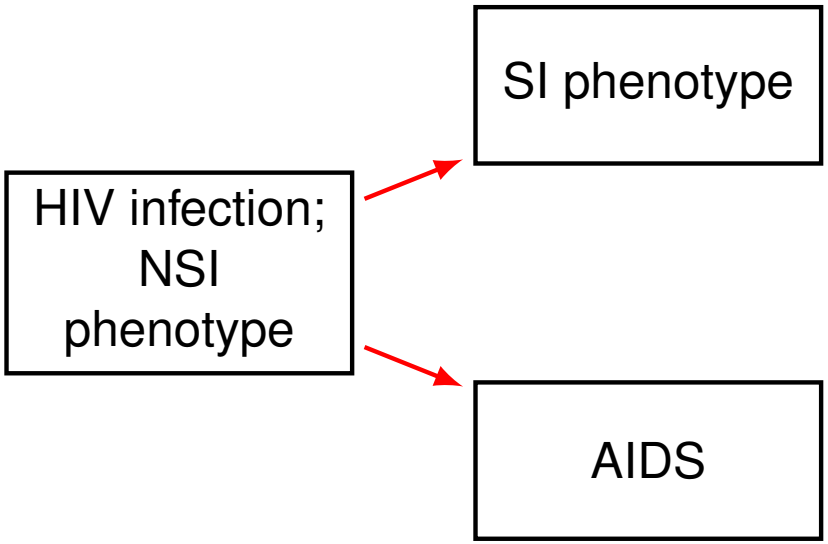
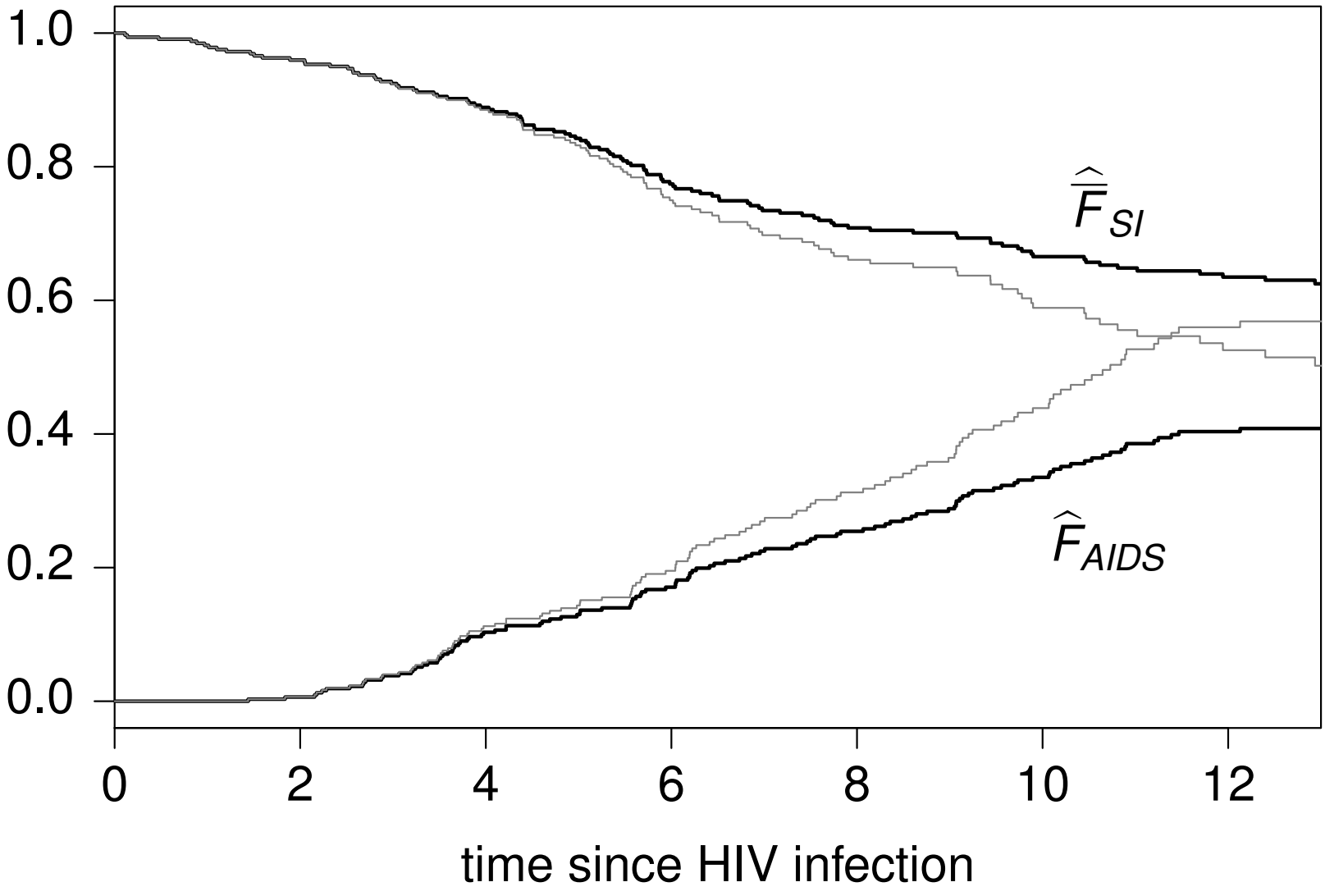
C: AIDS and SI switch after HIV infection

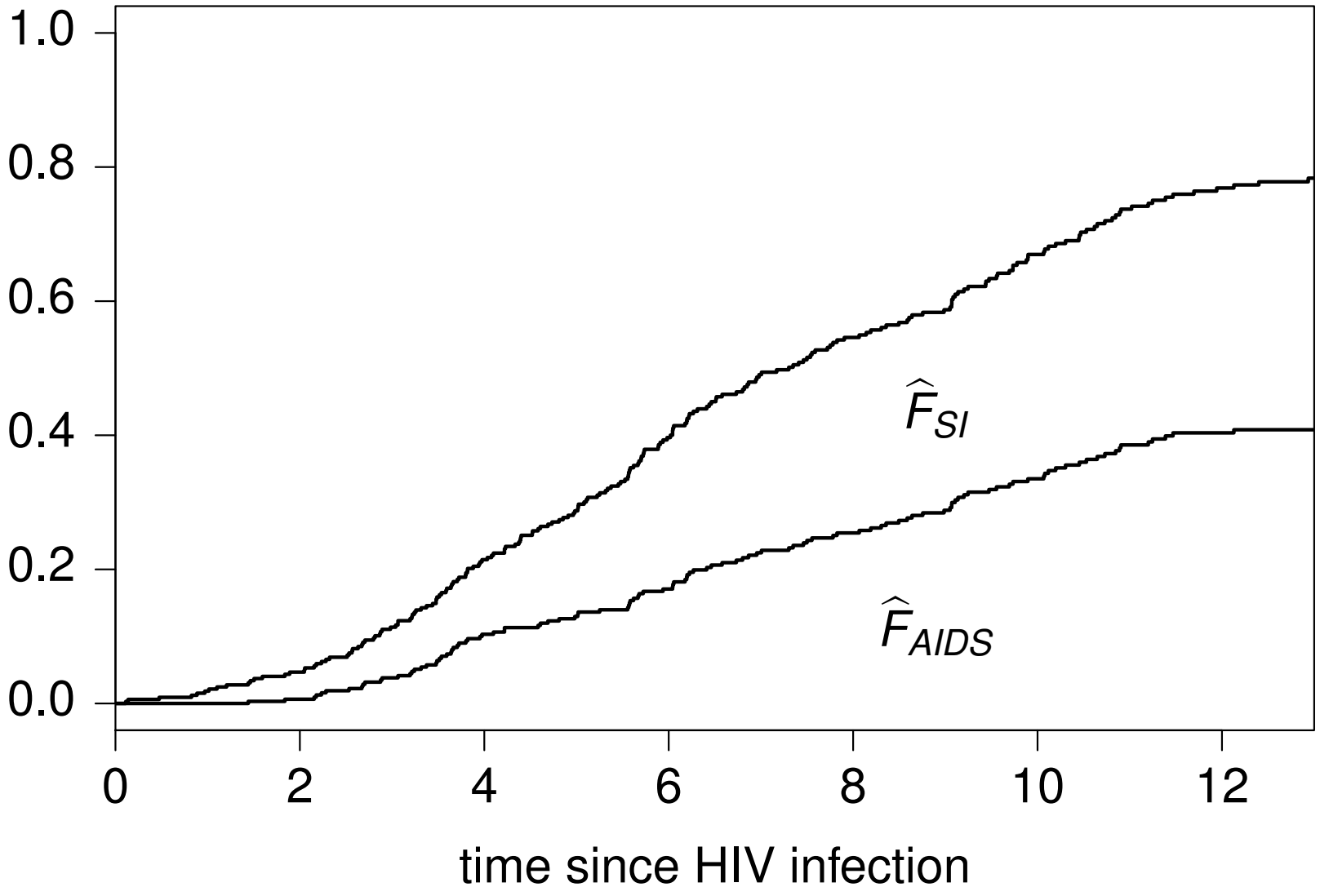


- SI: "syncytium inducing phenotype". At HIV infection only NSI
- Competing risks: time to either SI switch or AIDS as first event

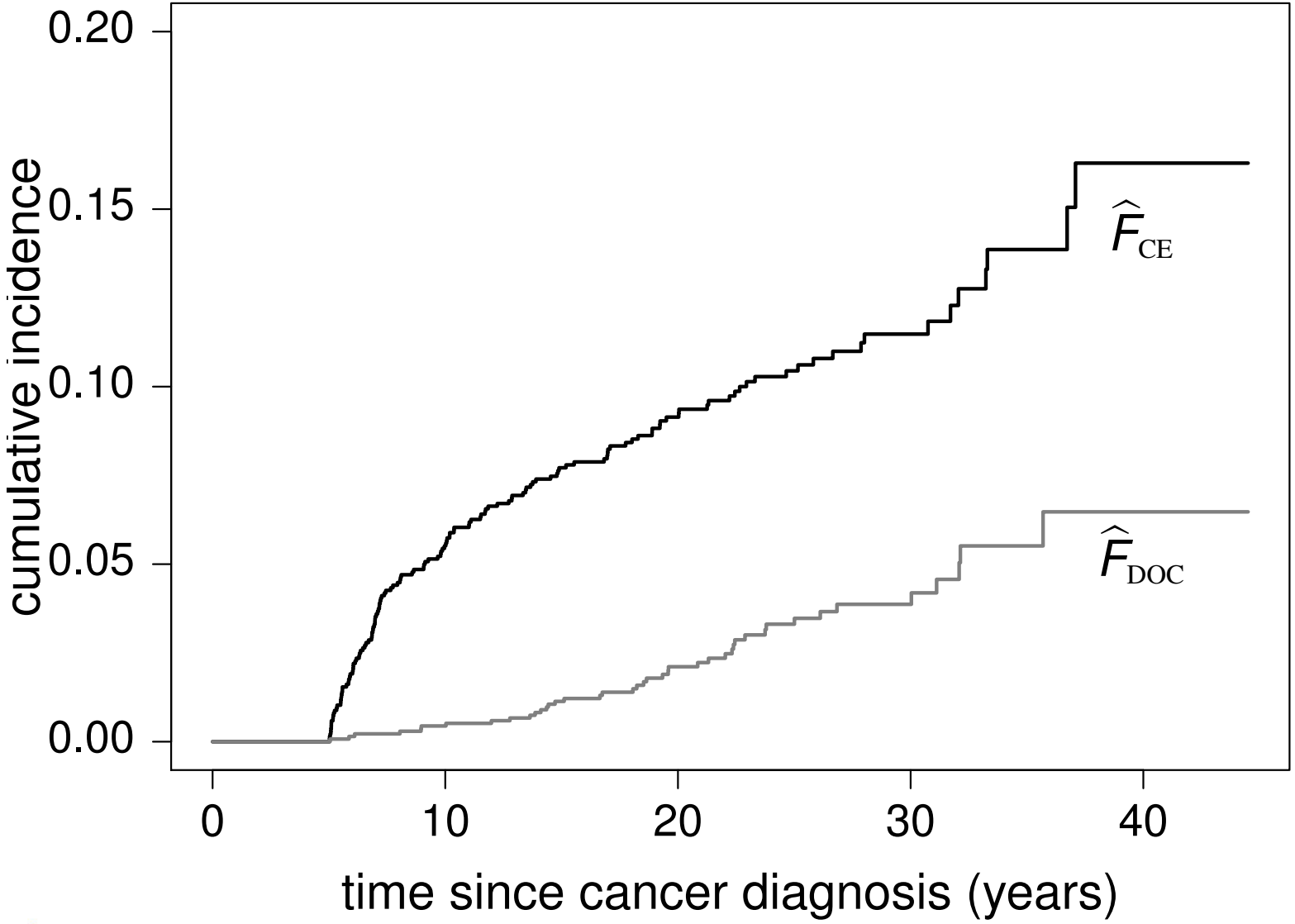
Alternate display format



Stacked display format

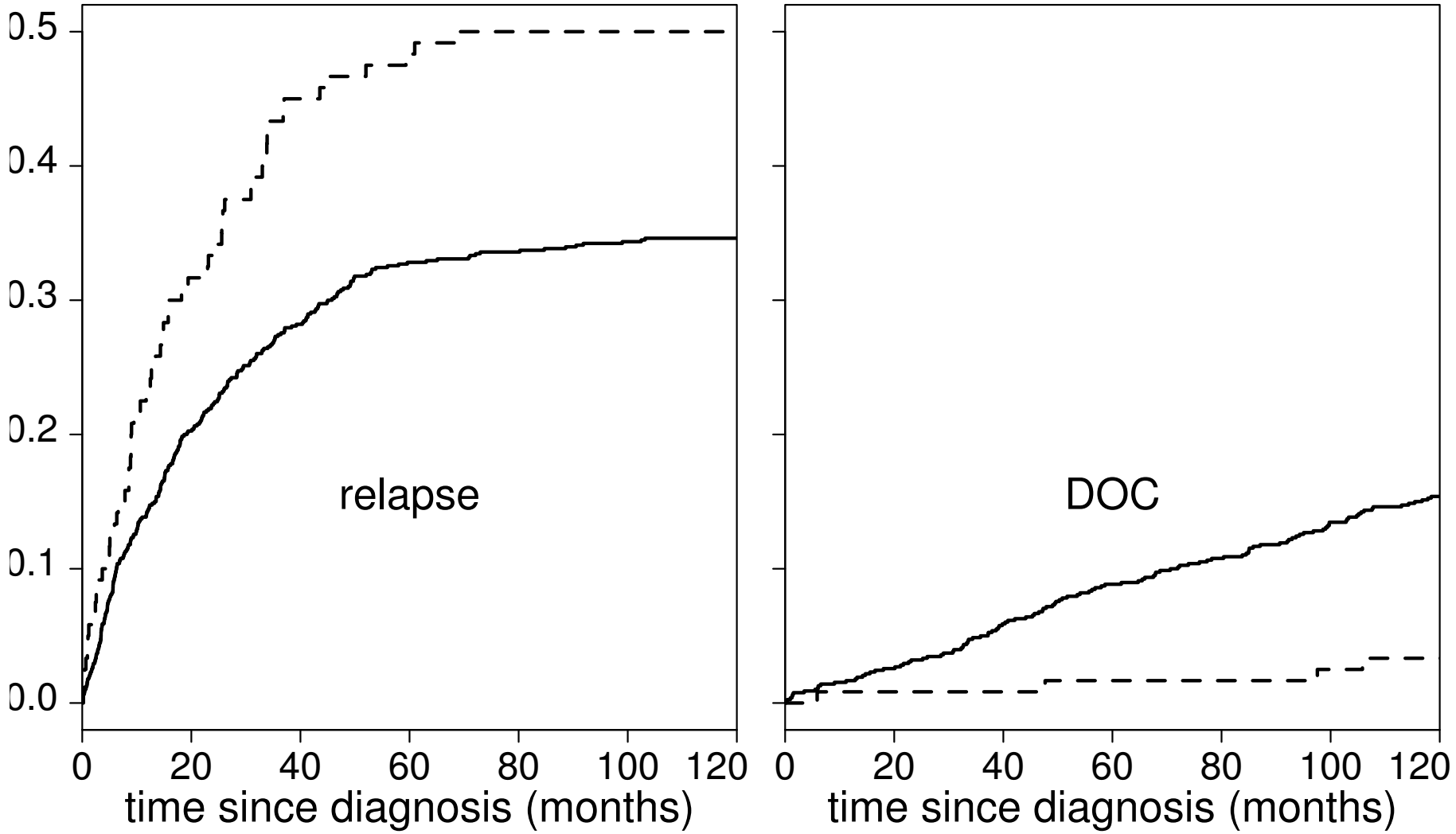


Overlaid display format (exercise 11.2)



Childhood cancer survivors; cardiac event, DOC competing

Separate display format, two groups (exercise 11.3)



Solid: male; dashed: female

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Rates and risks in competing risks setting

	hazard		cumulative	
competing risks	marginal	λ^m	net risk marginal survival function marginal cumulative incidence	$F^m(t)$
	cause-specific	λ_k	no corresponding quantity	
	subdistribution	h_k	crude risk cause-specific cumulative incidence	$F_k(t)$
combined	overall	h	overall risk overall survival function overall cumulative incidence	$F(t)$

Estimators in competing risks setting

hazard	estimate	cumulative
marginal	$d_k(t)/r(t)\{=\widehat{\lambda_k}(t)\}$	$\widehat{\overline{F^m}}^{\mathrm{PL}} = \prod_{t_{(j)}\le t}\left\{1-\widehat{\lambda_k}(t_{(j)})\right\}$
cause-specific	$\widehat{\lambda_k}(t) = d_k(t)/r(t)$	$\widehat{F_k}^{\mathrm{AJ}}(t) = \sum_{t_{(j)}\le t}\widehat{\overline{F}}^{\mathrm{PL}}(t_{(j)}-)\widehat{\lambda_k}(t_{(j)})$
subdistribution	$\widehat{h_k}(t) = d_k(t)/r^*(t)$	$\widehat{\overline{F_k}}^{\mathrm{PL}}(t) = \prod_{t_{(j)}\le t}\left\{1-\widehat{h_k}(t_{(j)})\right\}$
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Marginal distribution

- Estimated via (marginal) hazard, basis for Kaplan-Meier estimate of cumulative incidence/net risk
- Assumption: censoring (including from competing risk) is independent; censoring should give no information on residual time-to-event. Then censored individuals can be represented by the ones that remain at risk

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- Otherwise Kaplan-Meier has no interpretation.
  Does not describe survival in (hypothetical) world with competing event removed
- Extra information may allow to show dependent censoring (Example H-2, IDU and pre-AIDS death; Section 8.3), but independence can never be tested for

Estimators in competing risks setting

hazard	estimate	cumulative
marginal	$d_k(t)/r(t)\{=\widehat{\lambda_k}(t)\}$	$\widehat{F^m}^{\mathrm{PL}} = \prod_{t_{(j)}\le t}\left\{1-\widehat{\lambda_k}(t_{(j)})\right\}$
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  - Individuals censored by competing event don't have to be represented by the ones that remain at risk.
    Other censoring (administrative/loss to follow-up) must be independent

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- Cause-specific hazard
  - Basis for Aalen-Johansen estimator of cause-specific cumulative incidence/crude risk
  - If censoring due to competing event is independent, then marginal and cause-specific hazard are equal. Cumulative quantities different: Kaplan-Meier versus Aalen-Johansen

Estimators in competing risks setting

hazard	estimate	cumulative
marginal	$d_k(t)/r(t)\{=\widehat{\lambda_k}(t)\}$	$\widehat{F^m}^{\mathrm{PL}} = \prod_{t_{(j)}\le t}\left\{1-\widehat{\lambda_k}(t_{(j)})\right\}$
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Assumptions for data analysis

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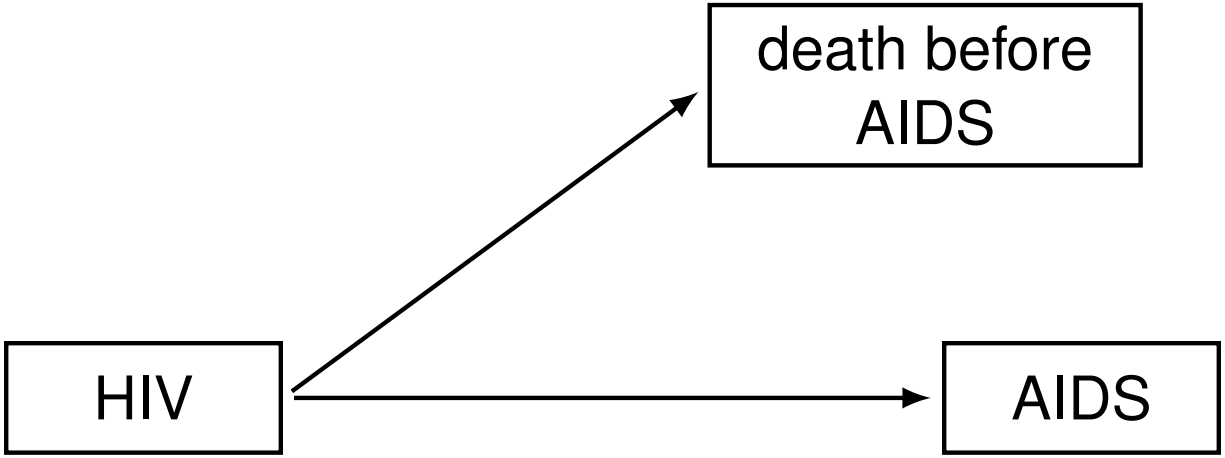
- Late entry non-informative

Time from HIV infection to AIDS



- Compare risk groups MSM and IDU
- IDUs expected to have faster progression to AIDS

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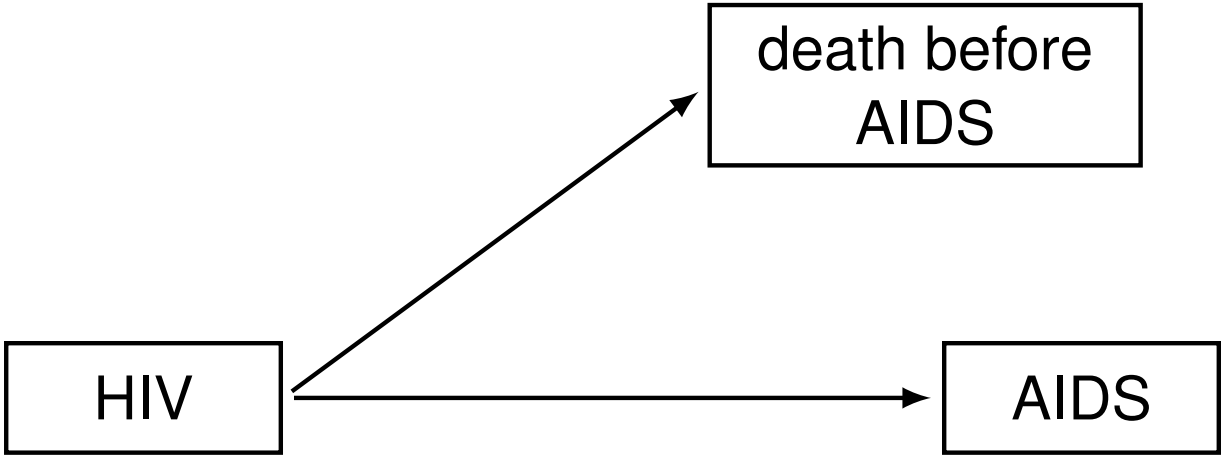

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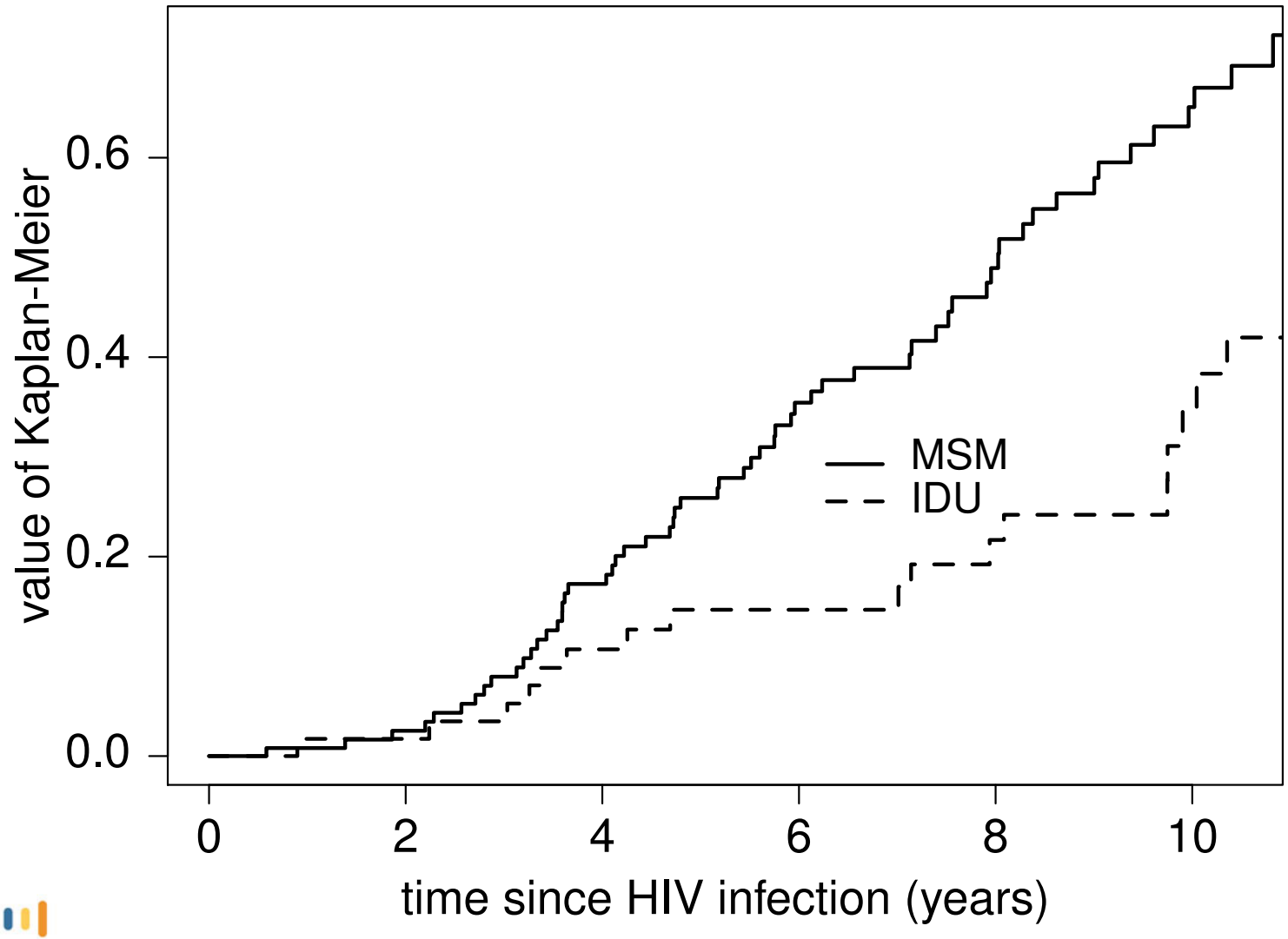
- Interest in time to AIDS if there were no pre-AIDS death.
  Interest in etiology and marginal distribution

Kaplan-Meier: IDU much slower progression ($p = 0.001$)



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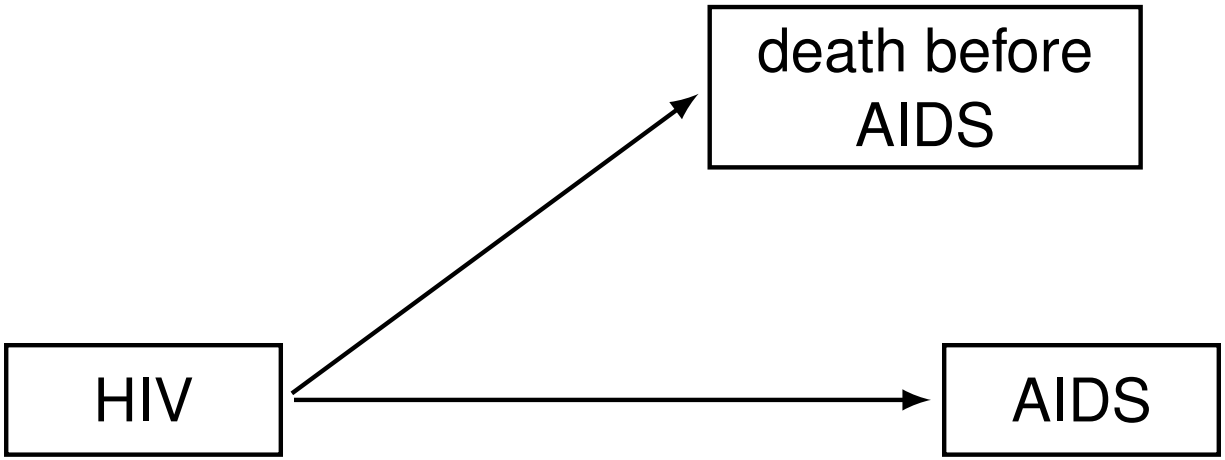

- Compare risk groups MSM and IDU
- IDUs expected to have faster progression to AIDS
- Interest in time to AIDS if there were no pre-AIDS death. Interest in etiology and marginal distribution
- Kaplan-Meier: those who die before AIDS leave the risk set Assumption: can be represented by those that do not die
- Can we test for independence?

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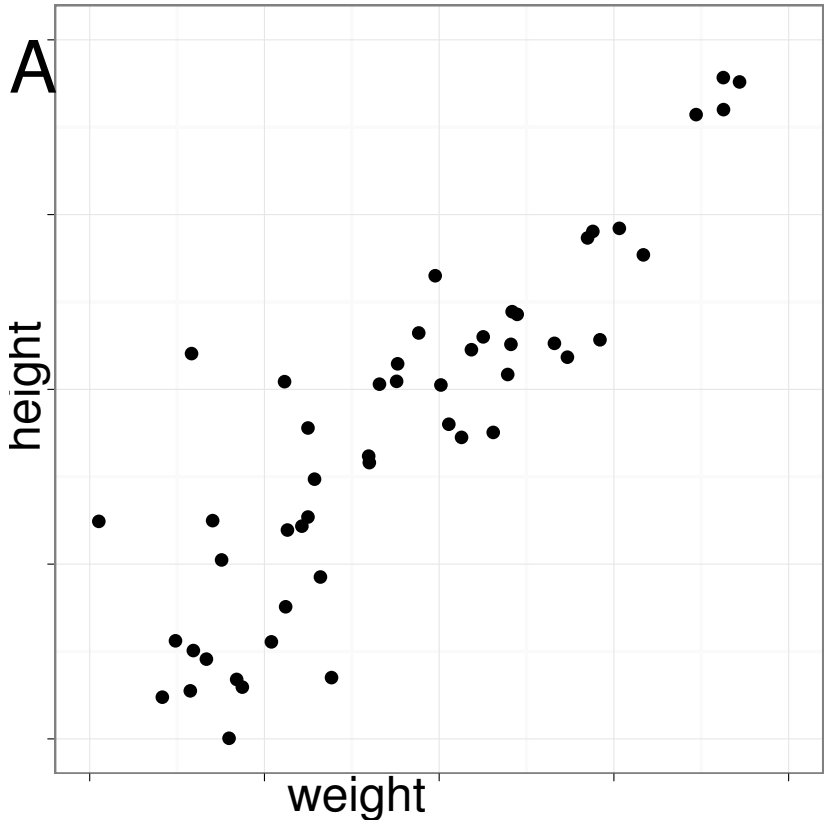
Cannot test for non-informative censoring



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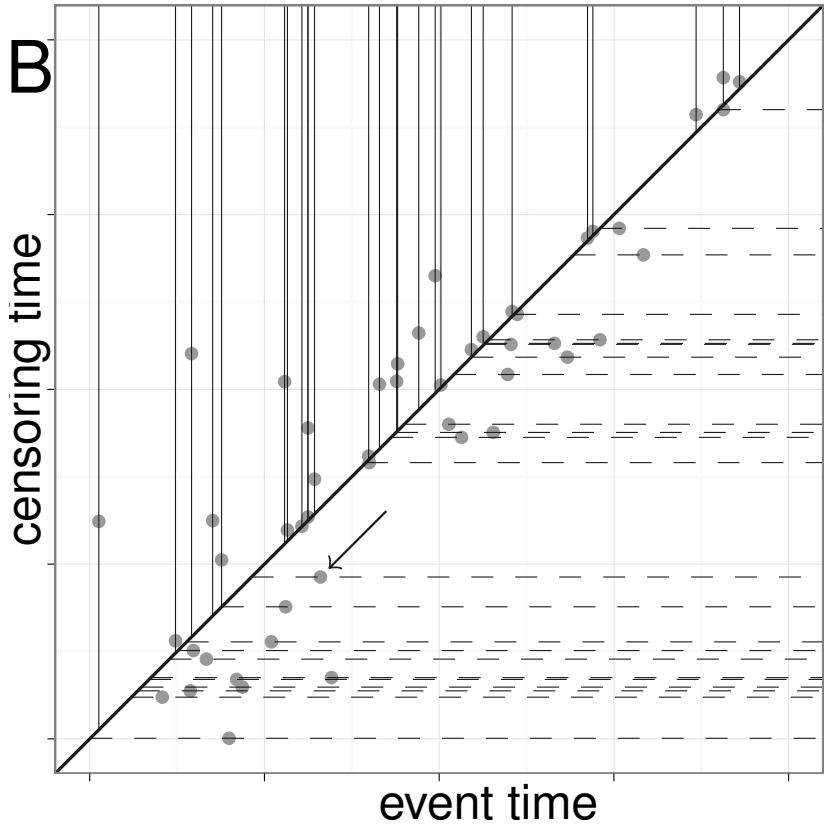

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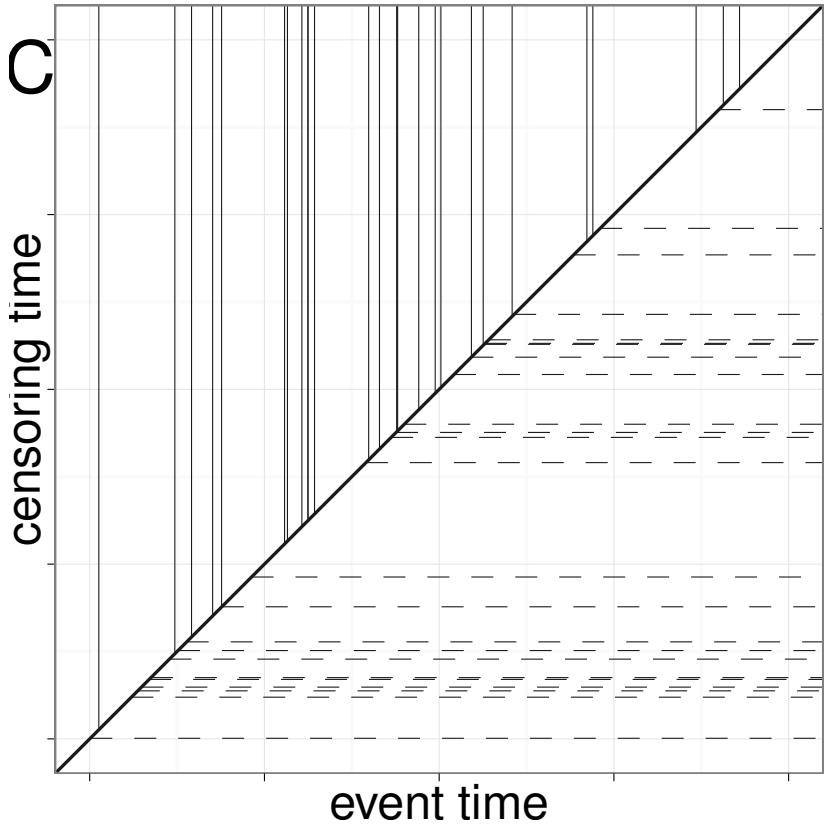

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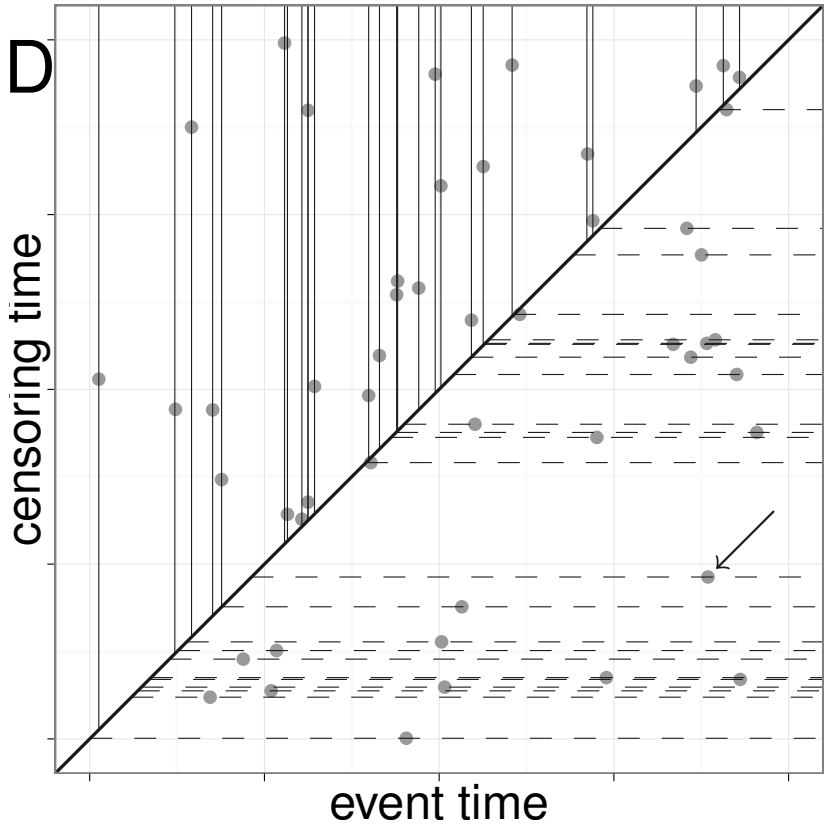

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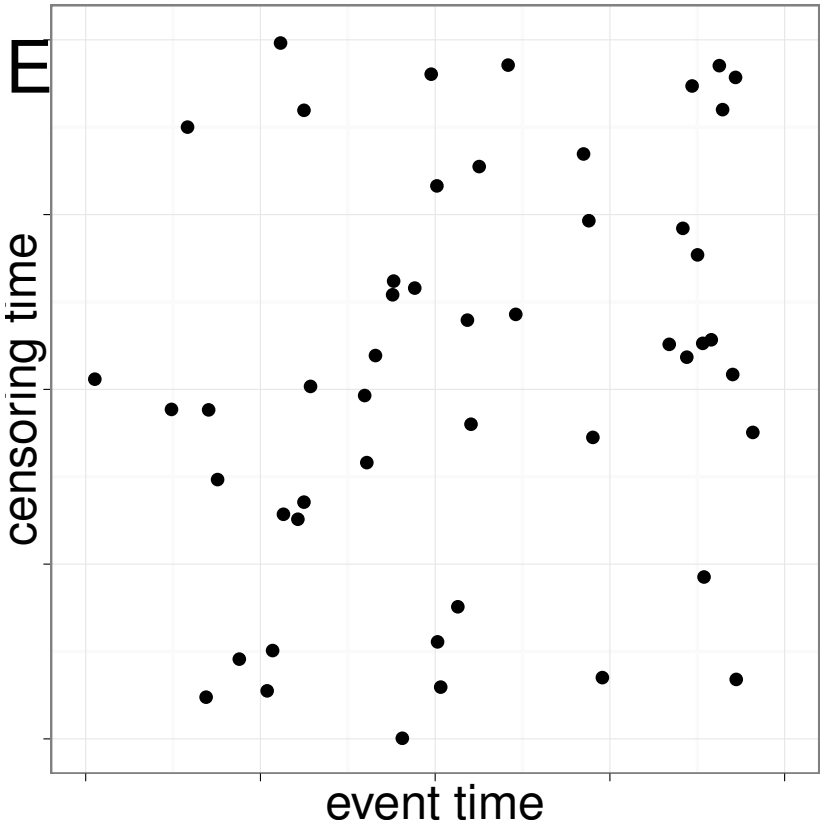

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Reasons for right censored data

Cutoff date of analysis (administrative censoring) Censoring usually non-informative

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Loss to follow-up May be problematic

- Sicker individuals discontinue participation in study (lack of energy, too ill, return to home country)
- Healthier individuals discontinue participation (don't feel the need to continue, start new life in other country)

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- Healthier individuals discontinue participation (don't feel the need to continue, start new life in other country)

Competing risks Often there is some background mechanism that influences progression to the different end points, which makes censoring informative. In competing risks analysis, independence is not required.

Exercise (part II).10.5: Don't we need to assume independence?

$$
\begin{aligned}
\overline{F}(t) &= \mathrm{P}(Y_1 > t, Y_2 > t, \ldots, Y_K > t) \\
&= \exp\left\{-\int_0^t h(s)ds\right\} \\
&= \exp\left\{-\sum_{e=1}^{K}\int_0^t \lambda_e(s)ds\right\} \\
&= \prod_{e=1}^{K}\exp\left\{-\int_0^t \lambda_e(s)ds\right\}
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$\exp\left\{-\int_0^t \lambda_e(s)ds\right\}$ estimated via $\exp\left\{-\sum_{j:t_{(j)}\leq t_{(i)}} \frac{d_e(t_{(j)})}{r(t_{(j)})}\right\}$ or

"Kaplan-Meier": $\prod_{j:t_{(j)}\leq t_{(i)}}\left(1 - \frac{d_e(t_{(j)})}{r(t_{(j)})}\right)$

Conclusion

- Same data can represent independent as well as dependent censoring
  - We cannot test for independence based on observed event/censoring data
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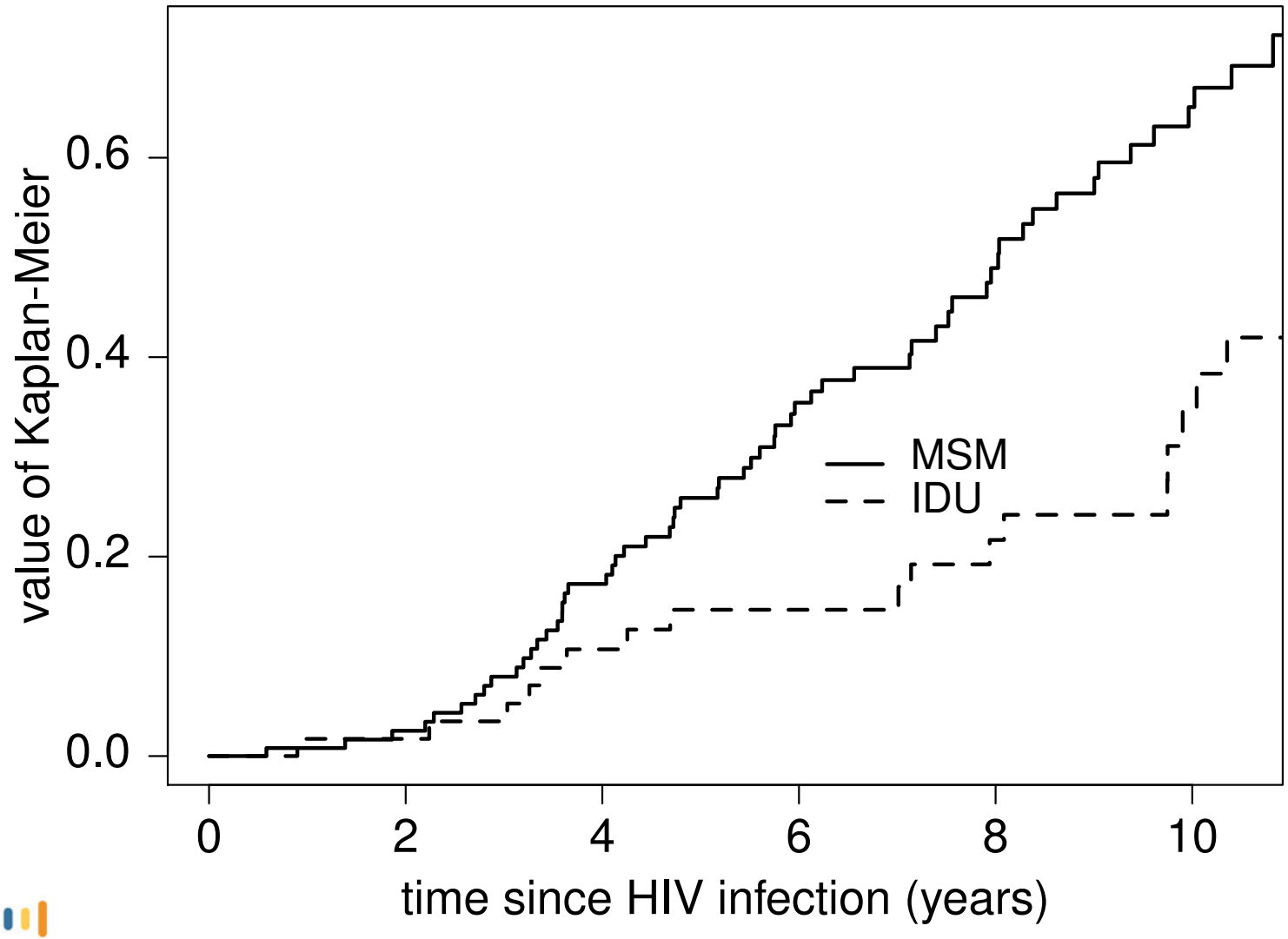
Explanation difference between IDU and MSM

- Extra information on cause of death before AIDS

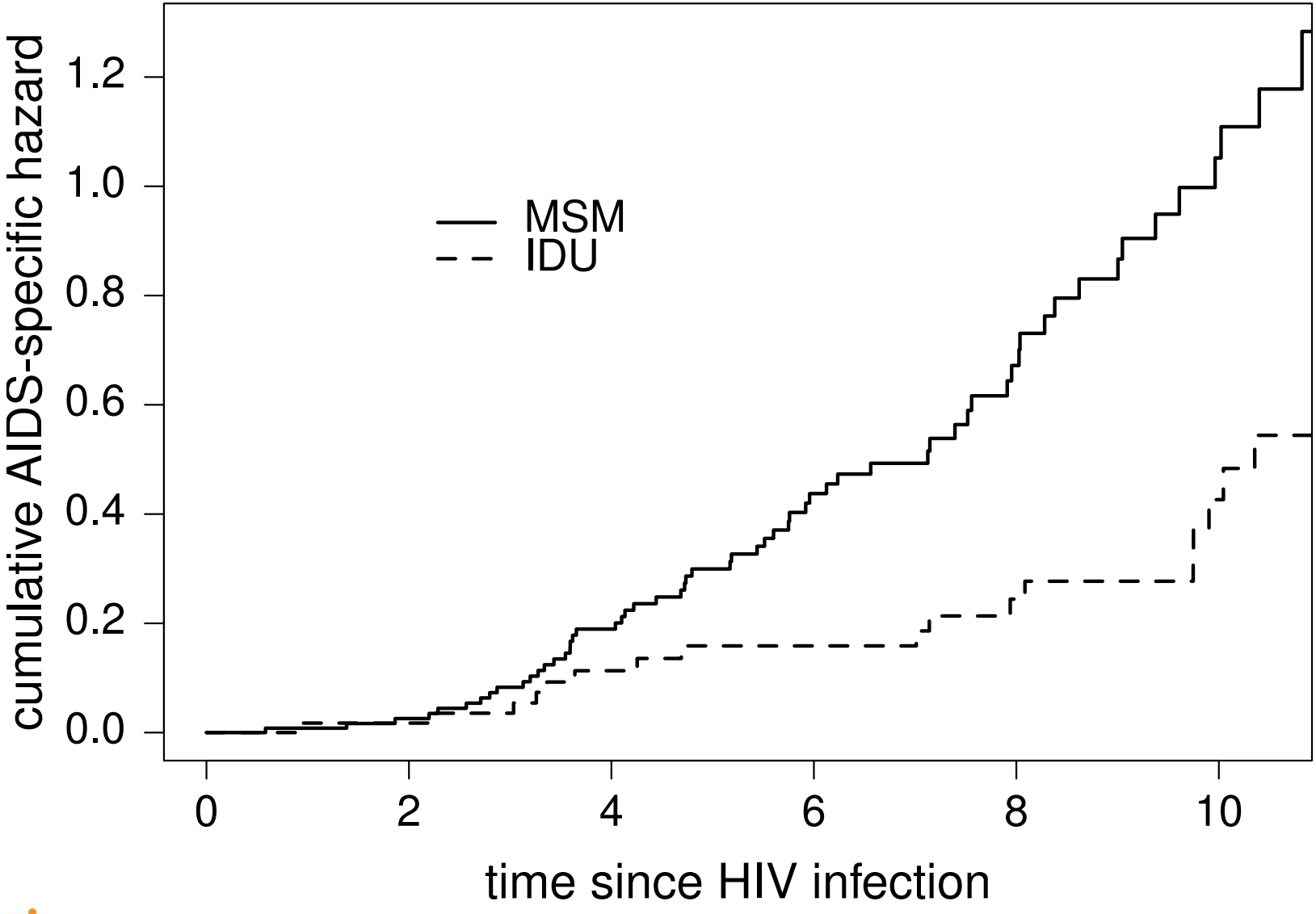
	IDU	MSM
Reason of death	Number	
HIV related infections	3	0
overdose/suicide	6	0
violence/accident	2	0
liver cirrhosis	2	0
cancer	0	1
heart attack	0	1
unknown	4	3

- Some pre-AIDS death causes in IDU related to AIDS progression. Censoring close to AIDS, hence net risk for IDU biased downwards

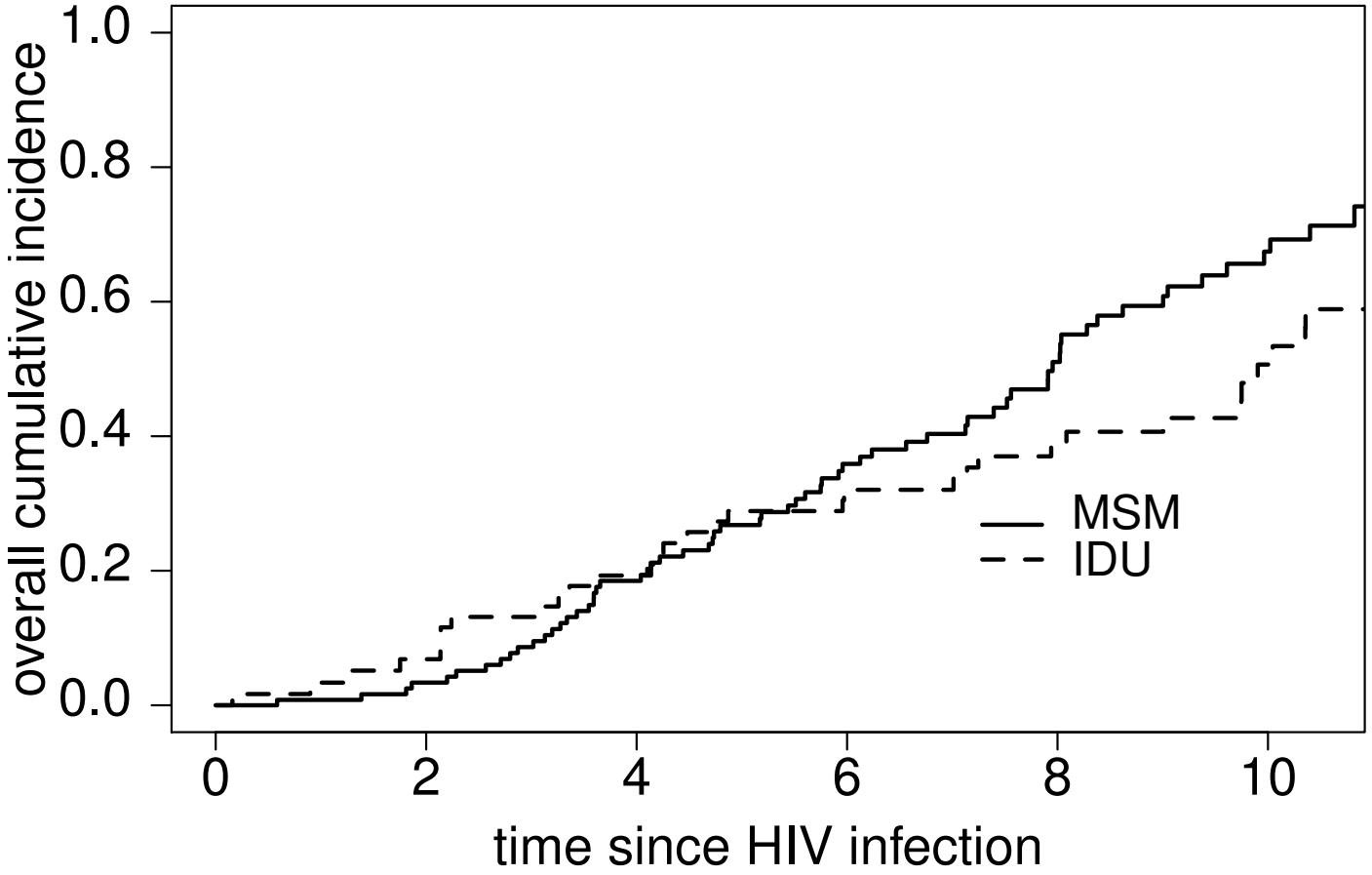
Kaplan-Meier: IDU much slower progression ($p = 0.001$)



AIDS-specific cumulative hazard

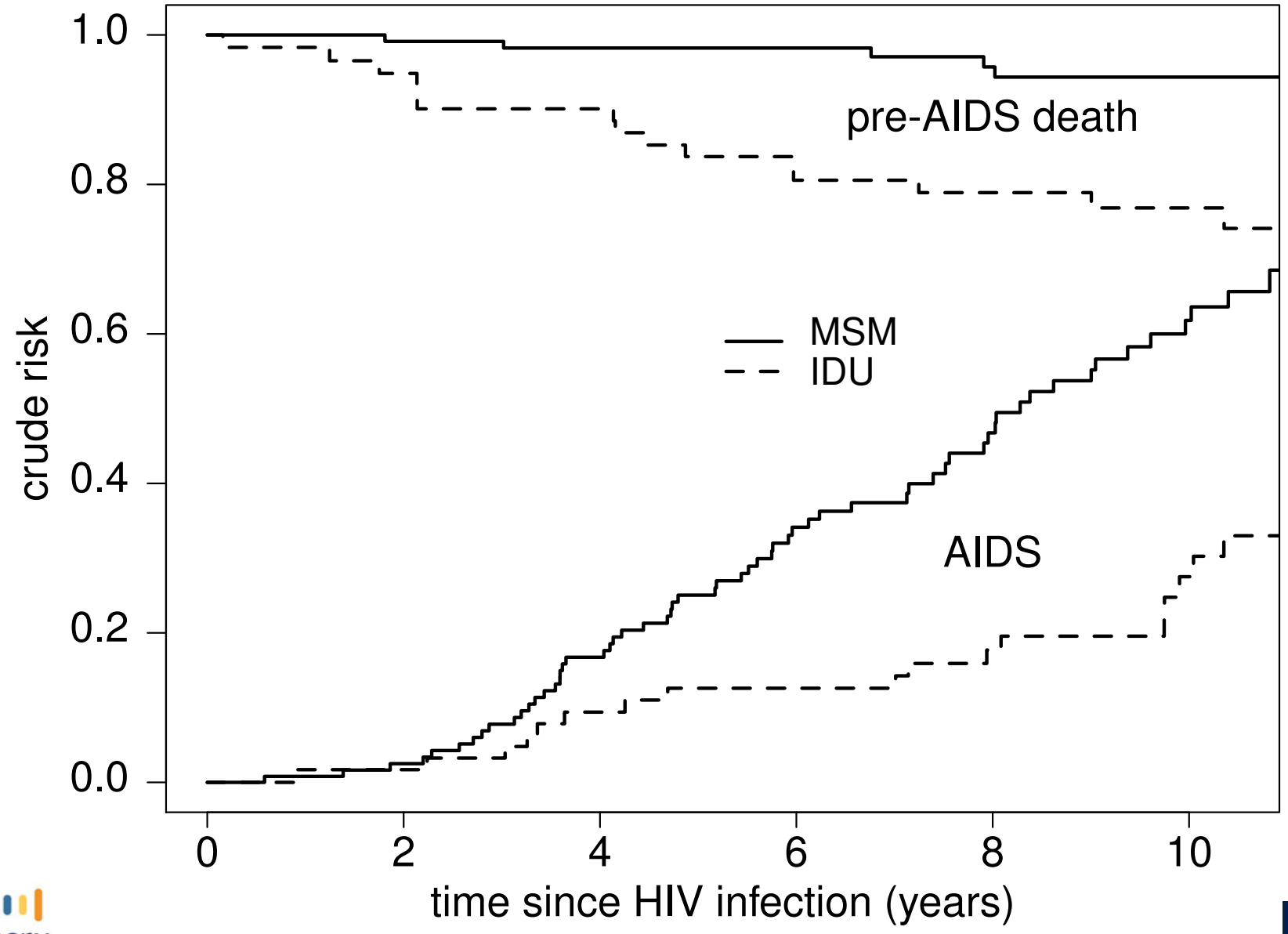


AIDS at death: combine AIDS and pre-AIDS death



Log-rank test: $p = 0.14$. Can we conclude that IDU and MSM have similar time to AIDS?

Never AIDS: death before AIDS treated as competing risk



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Observed data

$\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$
Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number "in risk set" (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \ldots, Z_{iq})$

Basic structure Cox model

$$h(t \mid \mathbf{Z}_i) = h_0(t)\ \exp(\boldsymbol{\beta}^\top \mathbf{Z}_i)$$

$h_0(t)$ baseline hazard, same for all individuals;
$\boldsymbol{\beta}^\top \mathbf{Z}_i = \beta_1 Z_{i1} + \ldots + \beta_q Z_{iq}$
Relative hazards constant over time:

$$\frac{h(t \mid \mathbf{Z}_i)}{h(t \mid \mathbf{Z}_j)} = \frac{\exp(\boldsymbol{\beta}^\top \mathbf{Z}_i)}{\exp(\boldsymbol{\beta}^\top \mathbf{Z}_j)}$$

Estimation

Estimate of β through maximisation of partial likelihood.

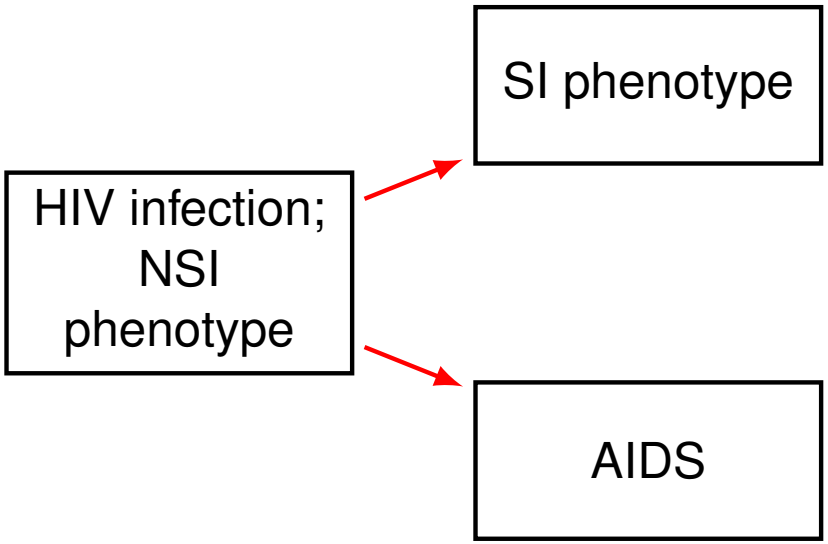
$$L(\beta) = \prod_{i=1,\delta_i=1}^{N} \left\{ \frac{\exp(\beta^\top \mathbf{Z}_i)}{\sum_{j=1}^{N} \psi_j(t_i) \exp(\beta^\top \mathbf{Z}_j)} \right\}.$$

with

$$\psi_j(t) = \begin{cases} 0 & \text{if person } j \text{ not at risk at } t \\ 1 & \text{if person } j \text{ is at risk at } t \end{cases}$$

- Only values at observed event times matter
- Numerator: $\mathbf{Z}_i$ from individual with event
- Denominator: all individuals in follow up at event time
- Directly extends to time-varying covariables $Z_i(t)$

C: AIDS and SI switch after HIV infection



- NSI virus needs CCR5 co-receptor for cell entry
- Gene coding for CCR5 may have deletion CCR5-Δ32 on one chromosome
- Individuals with CCR5-Δ32 have slower AIDS progression

- SI: "syncytium inducing phenotype". At HIV infection only NSI
- Competing risks: time to either SI switch or AIDS as first event

- Question: SI switch as first event more likely for CCR5-Δ32 individuals?

`aiddsi` data set, four example individuals

patnr	time	status	cause	ccr5
14	5.05	0	event-free	WW
3	2.23	1	AIDS	WW
15	10.20	1	AIDS	WM
8	8.61	2	SI	WW

Results Cox model for combined end point

```
coxph(formula = Surv(time, status != 0) ~ ccr5,
                                    data = aidssi)
```

Result:

```
         coef exp(coef) se(coef)     z       p
ccr5WM -0.701     0.496    0.186 -3.77 0.00016

Likelihood ratio test=16.5  on 1 df, p=4.95e-05  n= 324,
                        number of events= 220
   (5 observations deleted due to missingness)
```

Outline

Aalen-Johansen estimator
- Confidence intervals

Display formats

Marginal versus Cause-specific
- All quantities and their estimators
- Assumptions on censoring and delayed entry
- Independent right censoring?
- What can we do?

Regression; proportional cause-specific hazards
- Standard Cox model
- Proportional cause-specific hazards

Basic structure

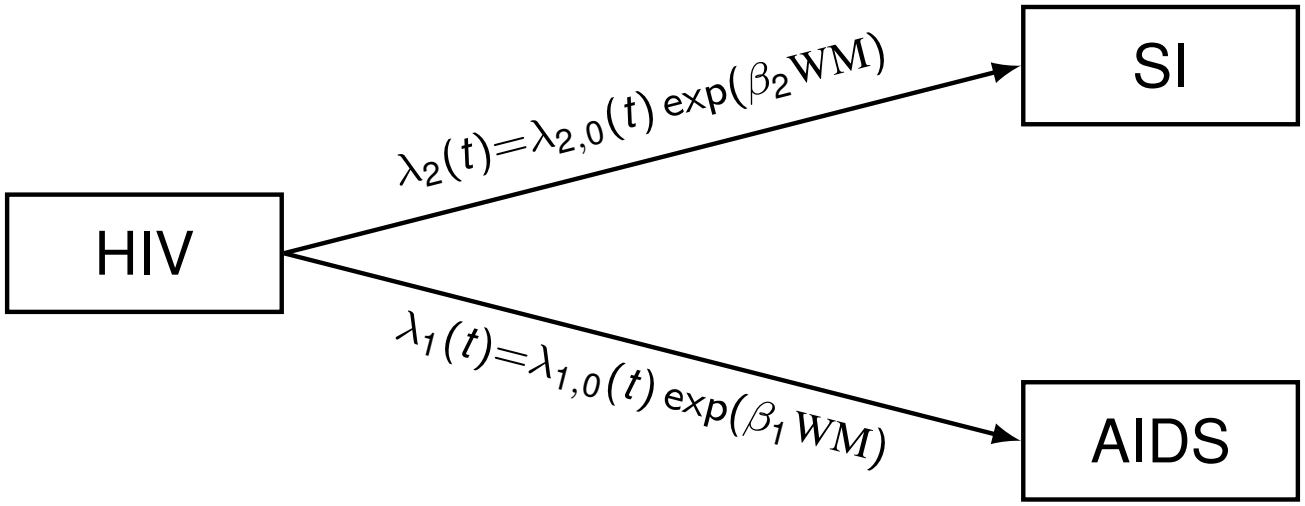
- $\lambda_k(t \mid \mathbf{Z}_i)$: cause-specific hazard of cause k for a subject with covariable vector $\mathbf{Z}_i$ given by

$$\lambda_k(t \,|\, \mathbf{Z}_i) = \lambda_{k,0}(t) \, \exp(\boldsymbol{\beta}_k^\top \mathbf{Z}_i) \; ,$$

with

  - $\lambda_{k,0}(t)$ is the baseline cause-specific hazard for cause k
  - $\boldsymbol{\beta}_k^\top = (\beta_{k1}, \ldots, \beta_{kq})$ represent the covariable effects on cause k

Example: effect CCR5Δ32 on AIDS and SI



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  with
  - $\lambda_{k,0}(t)$ is the baseline cause-specific hazard for cause k
  - $\boldsymbol{\beta}_k^\top = (\beta_{k1}, \ldots, \beta_{kq})$ represent the covariable effects on cause k
- Estimation via standard Cox model: individuals with a competing event leave the risk set

Analysis (in R)

```
coxph(Surv(time, status==1) ~ ccr5, data = aidssi)
------------------------
              coef exp(coef) se(coef)      z       p
ccr5WM.AIDS  -1.24     0.291    0.307  -4.02 5.7e-05

Likelihood ratio test=22  on 1 df, p=2.76e-06   n= 324,
                                   number of events= 113
   (5 observations deleted due to missingness)
```

Analysis (in R)

```
coxph(Surv(time, status==1) ~ ccr5, data = aidssi)
-------------------------
              coef exp(coef) se(coef)      z       p
ccr5WM.AIDS  -1.24     0.291    0.307  -4.02 5.7e-05

Likelihood ratio test=22  on 1 df, p=2.76e-06   n= 324,
                                   number of events= 113
   (5 observations deleted due to missingness)
```

```
coxph(Surv(time, status==2) ~ ccr5, data = aidssi)
-------------------------
            coef exp(coef) se(coef)     z       p
ccr5WM.SI -0.254     0.776    0.238 -1.07    0.29

Likelihood ratio test=1.19  on 1 df, p=0.275  n= 324,
                                   number of events= 107
   (5 observations deleted due to missingness)
```

Exercise 10.2

In a study on long-term effects of treatment in childhood cancer survivors, the end point of interest was symptomatic cardiac event (CE). Comment on the following statement:

> *Some individuals died without CE (DOC), but they had already left the study before they died. Therefore, for CE as end point, no competing risk of DOC is present in our data and the Kaplan-Meier curve is a valid estimate of the marginal distribution of time to CE.*

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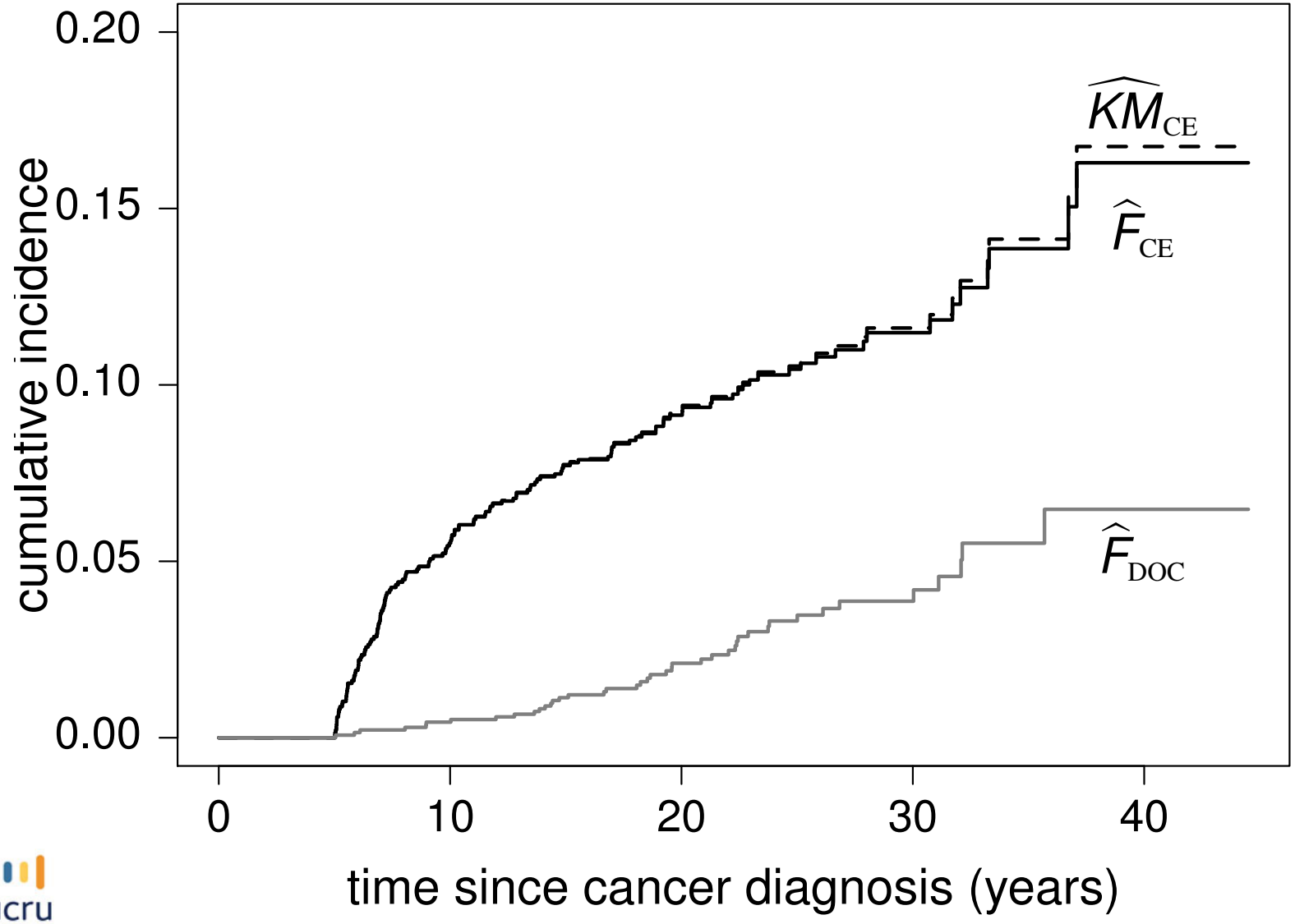
The Kaplan-Meier estimates the marginal distribution if all censoring is non-informative. Removing individuals from the risk set before they experience the competing event removes the competing risk, but does not solve the problem of informative censoring.

Exercise 10.3: Childhood cancer survivors; cardiac event, DOC competing



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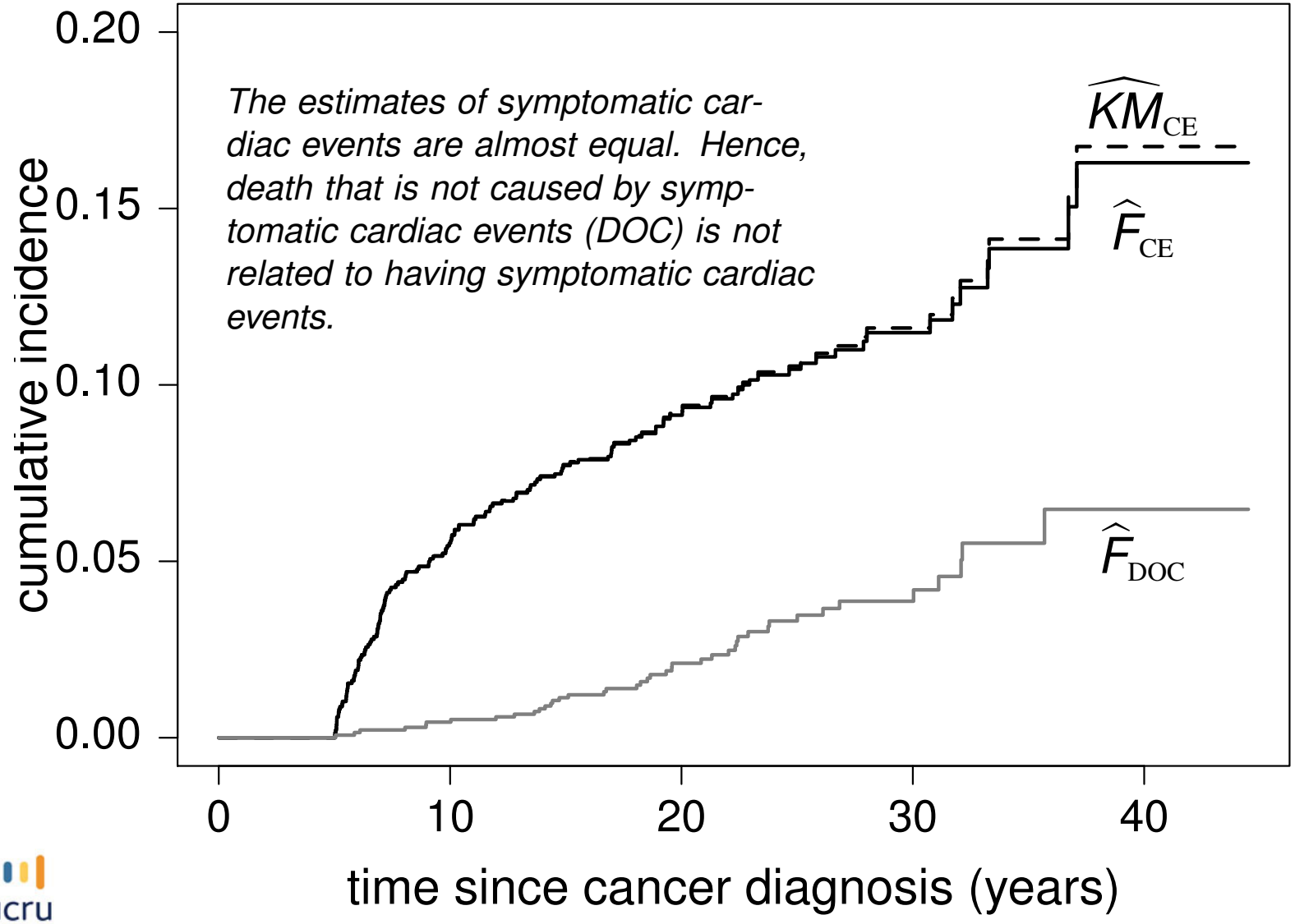
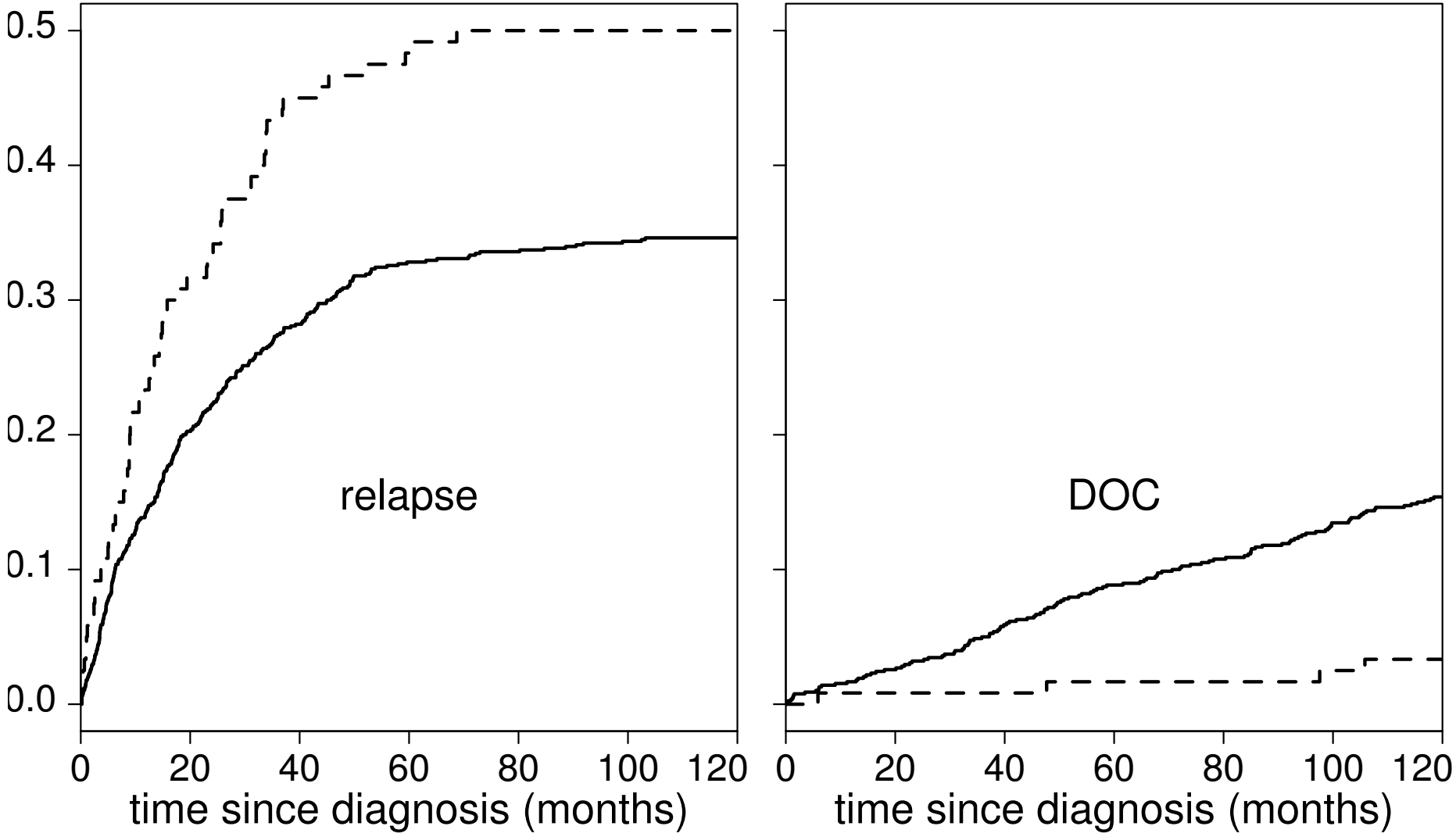


Answer

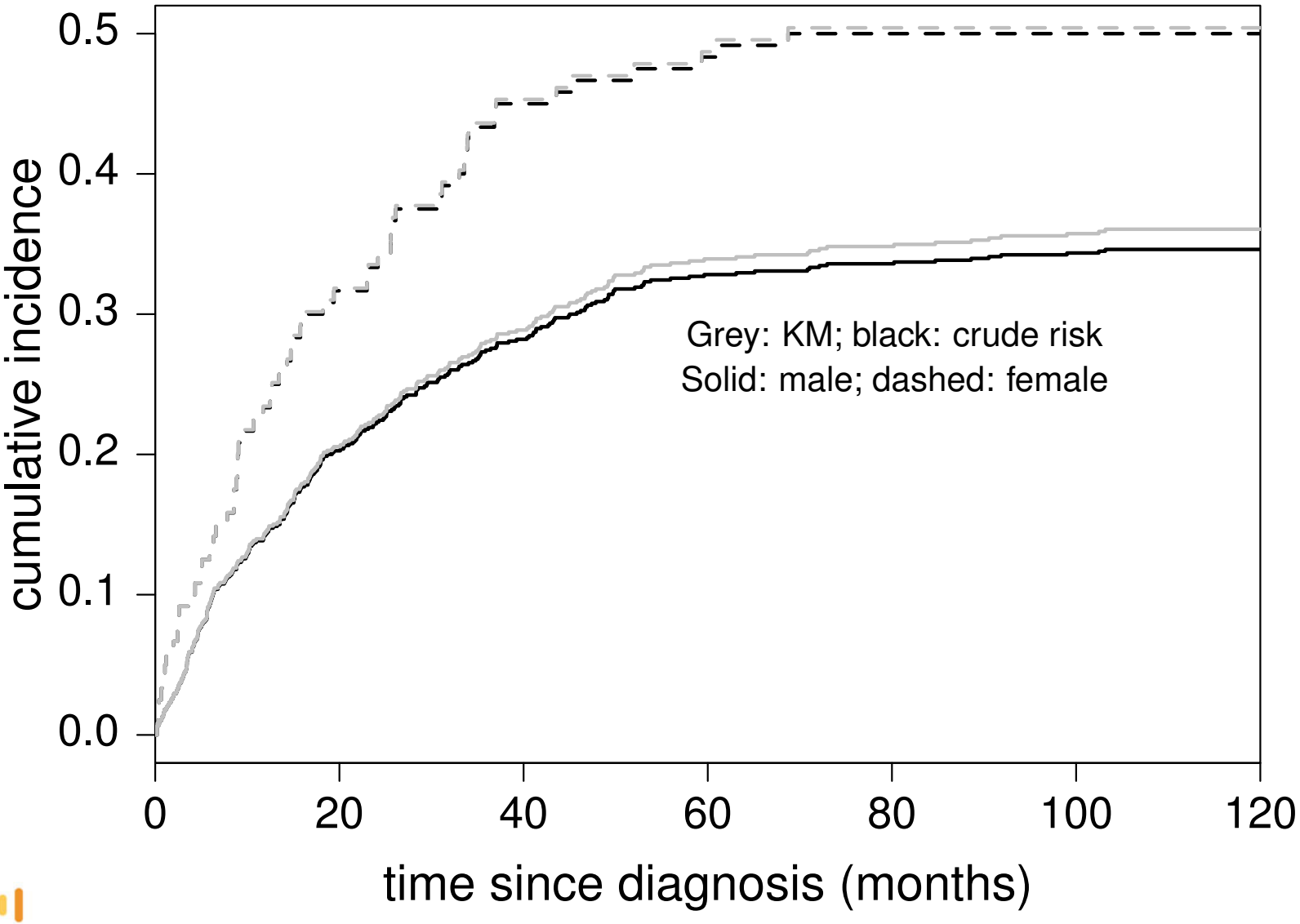
- The Kaplan-Meier and the estimator of the CE-specific cumulative incidence try to estimate different quantities
- Both curves are similar because there is little mortality due to other causes, at least during the first 20 years, when most of the CE's occur.
- Note that on one hand it is said that death due to other causes may not be related to CE events, whereas on the other hand it is called "informative censoring".

Exercise 10.4: Bladder cancer; relapse, DOC competing

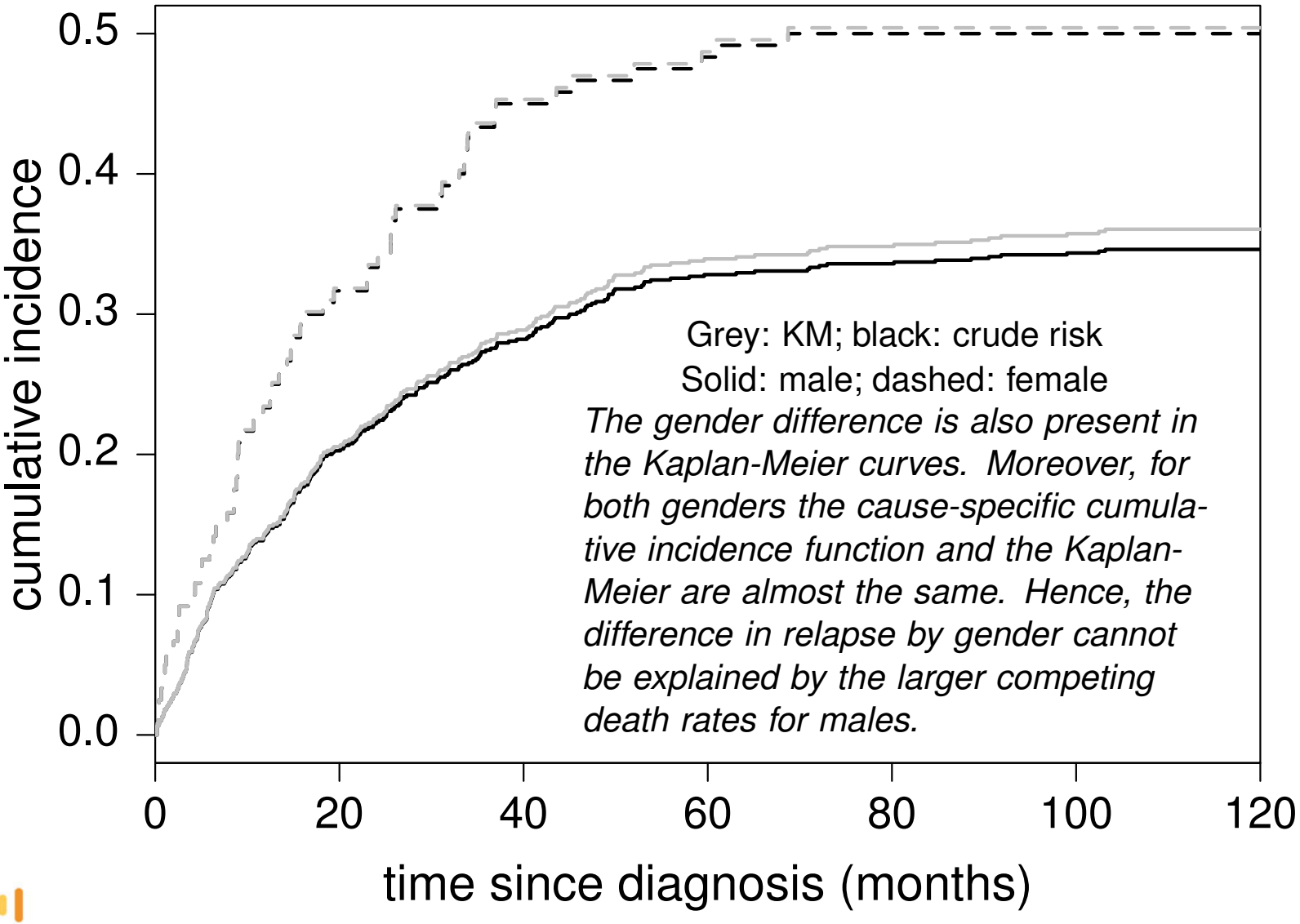


Left: relapse. Solid: male; dashed: female

Bladder cancer; relapse, DOC competing



Bladder cancer; relapse, DOC competing



The gender difference is also present in the Kaplan-Meier curves. Moreover, for both genders the cause-specific cumulative incidence function and the Kaplan-Meier are almost the same. Hence, the difference in relapse by gender cannot be explained by the larger competing death rates for males.

Answer

- The Kaplan-Meier tries to compare the marginal distribution of time to relapse for males and females. Only valid if DOC is noninformative for relapse.
- Estimates almost equal because there is little mortality due to other causes, at least during the first 40 months.

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Answer

- The Kaplan-Meier tries to compare the marginal distribution of time to relapse for males and females. Only valid if DOC is noninformative for relapse.
- Estimates almost equal because there is little mortality due to other causes, at least during the first 40 months.
- If we combine both event times, the curves for males and females will become similar. Would estimate marginal hazard if every person that died would have relapsed on the next day.
- All we can conclude is that females have a higher relapse-specific hazard than males. And females have a lower DOC-specific hazard than males.